

# The Lambert $W$ Function

A Real-Variable Guide to Its Branches, Identities, Series, Asymptotics, Bounds, Approximations, and Computation

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## Abstract

The Lambert  $W$  function is the inverse of the elementary-looking map  $w \mapsto we^w$ . It is the natural language for equations in which an unknown occurs both outside and inside an exponential. This article develops the function from first principles, at a level intended to be accessible after a standard sophomore course in calculus and a first course in proof. Special attention is paid to the two real branches: the principal branch  $W_0$  and the nonprincipal branch  $W_{-1}$ . We prove their domains, ranges, monotonicity, curvature, derivative formulas, integral identities, power series, square-root expansions at the common branch point, complete logarithmic asymptotic expansions, rigorous elementary bounds, and branch-safe Newton iterations. We also derive rational and continued fraction approximations, contour and real integral representations, and closed forms for representative transcendental equations.

The supplied source archive contains the foundational paper of Corless, Gonnet, Hare, Jeffrey, and Knuth [5], the NIST Digital Library of Mathematical Functions entry [7], papers on analytic approximations, continued fractions, certified high-precision evaluation, bounds for  $W_{-1}$ , Schroeder-type series, and applications in circuits, biochemical kinetics, ecology, and prime asymptotics. Those sources inform the presentation and bibliography. All principal results used in the exposition are proved here. Results that are mentioned only as historical or bibliographical context are explicitly labelled as such. A fully commented Python program supplied with the article reproduces the numerical figures.

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# 1 Why a new function is needed

Elementary algebra teaches us how to undo addition, multiplication, integer powers, and exponentials. Difficulty appears when the unknown occupies two different roles at once. For example,

$$ye^y = 7, \quad e^{3t} = 5t + 2, \quad s + \log s = 10$$

cannot generally be solved by a finite combination of the usual elementary functions. The first equation is so basic that it is sensible to name its inverse.

**Definition 1.1** (Lambert  $W$  function). *A value  $w$  of the Lambert  $W$  function at  $z$  is any solution of*

$$we^w = z. \tag{1}$$

Thus

$$w = W(z) \iff we^w = z.$$

Because the complex exponential is periodic and the map  $w \mapsto we^w$  is not one-to-one,  $W$  has branches. The branches are denoted  $W_k$ , where  $k \in \mathbb{Z}$ . On the real line only  $W_0$  and  $W_{-1}$  can take real values.

The symbol  $W$  is not an unexplained trick: it records an inverse operation, just as  $\sqrt{\phantom{x}}$ ,  $\log$ , and  $\arcsin$  do. Once the branch has been selected, every manipulation must be checked against (1). This simple habit prevents most branch errors.

## 1.1 A first numerical example

The equation  $we^w = 1$  has a unique real solution

$$\Omega := W_0(1) = 0.567143290409783872999968\dots,$$

sometimes called the *omega constant*. Indeed,  $\Omega e^\Omega = 1$ , or equivalently  $\Omega = e^{-\Omega}$ . The number is not mysterious once one sees it as the fixed point of  $w \mapsto e^{-w}$ .

## 1.2 Reading guide and prerequisites

Sections 2–5 require only calculus. The power-series sections use coefficient comparison and a proved version of Lagrange inversion. The asymptotic section introduces unsigned Stirling numbers of the first kind and proves the coefficient formula. The integral representation section uses a short, self-contained application of residues; it can be skipped without affecting the real-variable narrative. Numerical methods are developed from the mean value theorem and elementary logarithm inequalities.

Throughout,  $\log$  means the real natural logarithm when its argument is positive. The complex principal logarithm is written  $\text{Log}$ . At the point  $-1/e$ , both real branches equal  $-1$ . The notation  $W_{-1}(0) = -\infty$  is occasionally useful as an *extended* endpoint convention, but  $W_{-1}(0)$  is not a finite real number.

## 2 The two real branches

Everything begins with the graph of

$$f(w) = we^w.$$

Its derivative and second derivative are

$$f'(w) = e^w(1 + w), \quad f''(w) = e^w(2 + w).$$

The derivative vanishes only at  $w = -1$ , where

$$f(-1) = -\frac{1}{e}.$$

**Theorem 2.1** (Real branches, domains, and ranges). *The real solutions of  $we^w = x$  are as follows.*

- (i) *If  $x < -1/e$ , there is no real solution.*
- (ii) *If  $x = -1/e$ , there is one real solution,  $w = -1$ .*
- (iii) *If  $-1/e < x < 0$ , there are exactly two real solutions:*

$$W_0(x) \in (-1, 0), \quad W_{-1}(x) \in (-\infty, -1).$$

- (iv) *If  $x \geq 0$ , there is exactly one real solution,  $W_0(x) \in [0, \infty)$ .*

*More precisely,*

$$W_0 : [-1/e, \infty) \longrightarrow [-1, \infty)$$

*is a strictly increasing bijection, while*

$$W_{-1} : [-1/e, 0) \longrightarrow (-\infty, -1]$$

*is a strictly decreasing bijection.*

*Proof.* Because  $e^w > 0$ , the sign of  $f(w) = we^w$  is the sign of  $w$ . Moreover,

$$f'(w) < 0 \quad (w < -1), \quad f'(w) > 0 \quad (w > -1).$$

Thus  $f$  is strictly decreasing on  $(-\infty, -1]$  and strictly increasing on  $[-1, \infty)$ . The endpoint limits are

$$\lim_{w \rightarrow -\infty} we^w = 0^-, \quad f(-1) = -1/e, \quad f(0) = 0, \quad \lim_{w \rightarrow \infty} we^w = \infty.$$

Consequently the restriction of  $f$  to  $[-1, \infty)$  maps that interval bijectively onto  $[-1/e, \infty)$ , and its inverse is  $W_0$ . The restriction to  $(-\infty, -1]$  maps that interval bijectively, with reversed orientation, onto  $[-1/e, 0)$ , and its inverse is  $W_{-1}$ . The listed numbers of solutions follow immediately.  $\square$

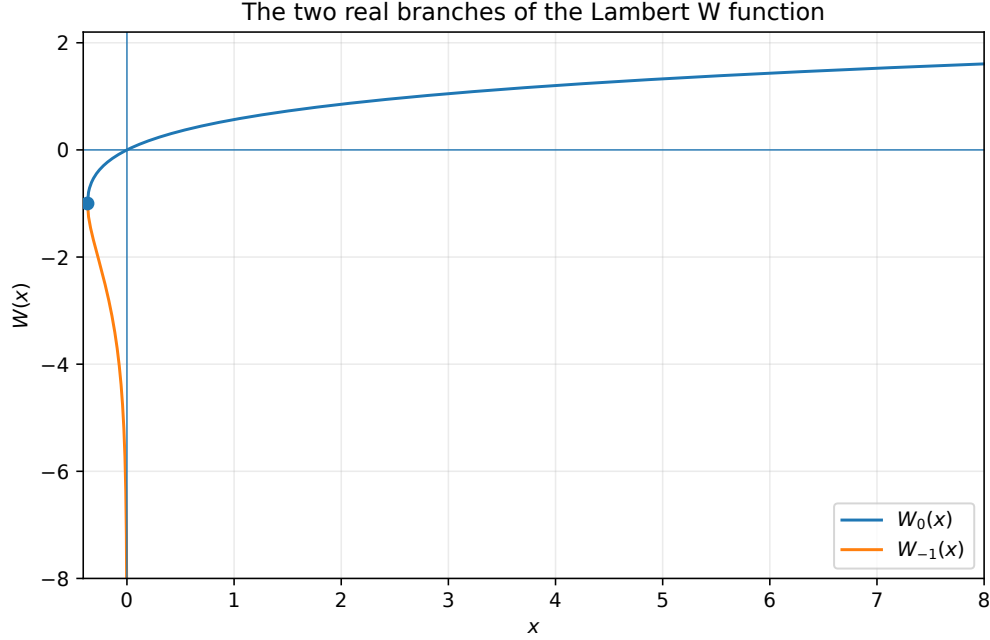


Figure 1: The principal branch  $W_0$  and the nonprincipal real branch  $W_{-1}$ . They meet at the square-root branch point  $(-1/e, -1)$ .

*Branch warning 2.2* (The subscript matters). For  $-1/e < x < 0$ , the notation  $W(x)$  is incomplete unless the branch is understood. For example,

$$W_0(-0.1) \approx -0.1118326, \quad W_{-1}(-0.1) \approx -3.5771521.$$

Both numbers satisfy  $we^w = -0.1$ .

**Corollary 2.3** (Branch-aware inverse identities). *For real  $a$ ,*

$$W_0(ae^a) = a \quad \text{when } a \geq -1,$$

*and*

$$W_{-1}(ae^a) = a \quad \text{when } a \leq -1.$$

*At  $a = -1$ , both formulas hold.*

*Proof.* The first statement says that  $W_0$  undoes the restriction of  $f(a) = ae^a$  to  $[-1, \infty)$ ; the second says the same about the restriction to  $(-\infty, -1]$ . This is precisely [Theorem 2.1](#).  $\square$

**Example 2.4** (A useful logarithmic identity). *Let  $x > 0$ . Taking  $a = \log x$  in [Theorem 2.3](#) gives*

$$W_0(x \log x) = \log x \quad \text{if } x \geq e^{-1},$$

*whereas*

$$W_{-1}(x \log x) = \log x \quad \text{if } 0 < x \leq e^{-1}.$$

*The same expression  $x \log x$  therefore selects different inverse branches on opposite sides of  $x = e^{-1}$ .*

### 3 Elementary identities and transformations

Let  $w = W_k(x)$  on any branch on which the displayed quantities are defined. The defining equation immediately yields

$$e^w = \frac{x}{w}, \quad e^{-w} = \frac{w}{x}, \quad \log |x| = w + \log |w| \quad (2)$$

for real  $x \neq 0$ . The last identity is often more stable numerically than  $we^w = x$ , because it avoids overflow and underflow.

**Proposition 3.1** (An addition identity). *Suppose  $a = W_j(x)$ ,  $b = W_\ell(y)$ , and  $a, b, ab \neq 0$ . Then*

$$(a + b)e^{a+b} = xy \left( \frac{1}{a} + \frac{1}{b} \right). \quad (3)$$

*Consequently*

$$a + b = W_m \left( xy \left( \frac{1}{a} + \frac{1}{b} \right) \right)$$

*for whichever branch  $m$  contains the value  $a + b$ .*

*Proof.* Since  $e^a = x/a$  and  $e^b = y/b$ ,

$$(a + b)e^{a+b} = (a + b) \frac{x}{a} \frac{y}{b} = xy \frac{a + b}{ab} = xy \left( \frac{1}{a} + \frac{1}{b} \right).$$

Applying the inverse branch containing  $a + b$  proves the second statement. □

The final branch qualification is essential. Identities involving an inverse function cannot determine a branch merely by algebra.

#### 3.1 The Wright omega viewpoint

For  $t \in \mathbb{R}$ , define

$$\omega(t) := W_0(e^t).$$

Then  $\omega(t) > 0$  and

$$\omega(t) + \log \omega(t) = t. \quad (4)$$

Conversely, a positive solution of (4) is  $\omega(t)$ . This reparametrization is valuable when  $e^t$  is too large to store.

There is an equally natural two-branch negative analogue. For  $u \geq 1$ , the equation

$$v - \log v = u \quad (5)$$

has two positive solutions, one in  $(0, 1]$  and one in  $[1, \infty)$ :

$$v_0(u) = -W_0(-e^{-u}), \quad v_1(u) = -W_{-1}(-e^{-u}).$$

*Proof.* Writing  $w = -v$  in  $we^w = -e^{-u}$  gives  $ve^{-v} = e^{-u}$ . Taking logarithms gives  $v - \log v = u$ . The function  $v - \log v$  decreases on  $(0, 1]$ , increases on  $[1, \infty)$ , and has minimum 1 at  $v = 1$ , so the two solutions and their locations follow. □

## 4 Exact relations between the two real branches

On the interval  $(-1/e, 0)$ , the two real branches are not independent. They are the two preimages of the same number under  $w \mapsto we^w$ . The following parametrization describes the pair exactly and, in particular, places Bernoulli numbers naturally into the theory.

**Theorem 4.1** (Exact branch-pair parametrization). *Let  $x \in (-1/e, 0)$ , and define the positive branch gap*

$$\Delta = W_0(x) - W_{-1}(x) > 0. \quad (6)$$

*Then*

$$W_0(x) = -\frac{\Delta}{e^\Delta - 1}, \quad (7)$$

$$W_{-1}(x) = -\frac{\Delta}{1 - e^{-\Delta}} = -\frac{\Delta e^\Delta}{e^\Delta - 1}, \quad (8)$$

$$x = -\frac{\Delta}{e^\Delta - 1} \exp\left(-\frac{\Delta}{e^\Delta - 1}\right). \quad (9)$$

*Conversely, every  $\Delta > 0$  inserted into these formulas produces one and only one  $x \in (-1/e, 0)$ , with the displayed values equal to  $W_0(x)$  and  $W_{-1}(x)$ .*

*Equivalently, if  $t = e^\Delta > 1$ , then*

$$W_0(x) = -\frac{\log t}{t - 1}, \quad W_{-1}(x) = -\frac{t \log t}{t - 1}, \quad (10)$$

$$x = -\frac{\log t}{t - 1} t^{-1/(t-1)}. \quad (11)$$

*Proof.* Put  $a = W_0(x)$  and  $b = W_{-1}(x)$ . Then  $a - b = \Delta$ , and the equality  $ae^a = be^b = x$  gives

$$\frac{a}{b} = e^{b-a} = e^{-\Delta}.$$

Thus  $a = be^{-\Delta}$ , and subtraction yields

$$\Delta = a - b = b(e^{-\Delta} - 1).$$

Solving first for  $b$ , and then for  $a$ , proves (7) and (8). Substitution into  $x = ae^a$  proves (9). Replacing  $e^\Delta$  by  $t$  gives (10) and (11).

It remains to justify the converse and the stated range of  $x$ . For  $\Delta > 0$ , the elementary inequality  $e^\Delta > 1 + \Delta$  gives

$$-1 < -\frac{\Delta}{e^\Delta - 1} < 0.$$

Also

$$-\frac{\Delta}{1 - e^{-\Delta}} < -1,$$

because  $1 - e^{-\Delta} < \Delta$ . The two displayed numbers have positive gap  $\Delta$ , and direct substitution shows that their products with their exponentials are equal. Their common value is therefore in  $(-1/e, 0)$  by Theorem 2.1, and the branch ranges identify them uniquely as  $W_0$  and  $W_{-1}$ . This also proves that every  $\Delta > 0$  gives exactly one paired input.  $\square$



**Corollary 4.2** (Exact symmetric identities). *For  $-1/e < x < 0$ , with  $\Delta$  as in (6),*

$$\frac{W_{-1}(x)}{W_0(x)} = e^\Delta, \quad (12)$$

$$W_0(x) + W_{-1}(x) = -\Delta \coth\left(\frac{\Delta}{2}\right), \quad (13)$$

$$W_0(x)W_{-1}(x) = \frac{\Delta^2}{4 \sinh^2(\Delta/2)}. \quad (14)$$

*In particular,*

$$W_0(x) + W_{-1}(x) < -2, \quad 0 < W_0(x)W_{-1}(x) < 1. \quad (15)$$

*Proof.* The ratio follows immediately from (7)–(8). Adding and multiplying those formulas, and using

$$\coth(y) = \frac{e^{2y} + 1}{e^{2y} - 1}, \quad 4 \sinh^2(y) = e^{2y} - 2 + e^{-2y},$$

gives (13) and (14).

For  $y > 0$ ,  $\sinh y > y$ , so the product in (14) lies strictly between 0 and 1. Also  $y \cosh y - \sinh y$  vanishes at 0 and has derivative  $y \sinh y > 0$ ; hence  $y \coth y > 1$ . Taking  $y = \Delta/2$  proves the sum inequality.  $\square$

#### 4.1 Bernoulli-number expansions of the paired branches

The Bernoulli numbers  $B_n$  are defined by the generating function

$$\frac{z}{e^z - 1} = \sum_{n=0}^{\infty} B_n \frac{z^n}{n!}, \quad |z| < 2\pi. \quad (16)$$

Thus  $B_0 = 1$ ,  $B_1 = -1/2$ ,  $B_2 = 1/6$ ,  $B_4 = -1/30$ , and all odd  $B_n$  beyond  $B_1$  vanish.

**Corollary 4.3** (Bernoulli expansion in the branch gap). *For  $0 < \Delta < 2\pi$ ,*

$$W_0(x) = -1 + \frac{\Delta}{2} - \frac{\Delta^2}{12} + \frac{\Delta^4}{720} - \frac{\Delta^6}{30240} + \cdots, \quad (17)$$

$$W_{-1}(x) = -1 - \frac{\Delta}{2} - \frac{\Delta^2}{12} + \frac{\Delta^4}{720} - \frac{\Delta^6}{30240} + \cdots. \quad (18)$$

*More compactly,*

$$W_0(x) = -\sum_{n=0}^{\infty} B_n \frac{\Delta^n}{n!}, \quad W_{-1}(x) = -\Delta - \sum_{n=0}^{\infty} B_n \frac{\Delta^n}{n!}. \quad (19)$$

*Proof.* The first formula in (19) is (7) together with (16). The identity

$$\frac{\Delta}{1 - e^{-\Delta}} = \Delta + \frac{\Delta}{e^\Delta - 1}$$

gives the second. Substitution of the first Bernoulli numbers gives the expanded forms. The convergence radius is  $2\pi$ , because the nearest nonzero zeros of  $e^z - 1$  are  $z = \pm 2\pi i$ .  $\square$

*Remark 4.4.* The gap variable  $\Delta$  is different from the square-root variable used near the branch point in Section 9. They are asymptotically equivalent up to a factor: by (53),  $\Delta \sim 2\sqrt{2(1 + ex)}$ . The square-root variable is best for evaluating a single branch near  $-1/e$ ; the gap variable is best for relations that involve both branches at once.

## 5 Differential calculus and geometry

### 5.1 First and second derivatives

**Theorem 5.1** (Derivative formulas). *At every real point where  $W_k(x) \neq -1$ ,*

$$W'_k(x) = \frac{1}{e^{W_k(x)}(1 + W_k(x))} = \frac{W_k(x)}{x(1 + W_k(x))}, \quad (20)$$

$$W''_k(x) = -\frac{e^{-2W_k(x)}(W_k(x) + 2)}{(1 + W_k(x))^3} \quad (21)$$

$$= -\frac{W_k(x)^2(W_k(x) + 2)}{x^2(1 + W_k(x))^3}. \quad (22)$$

*The second forms are interpreted by limits at  $x = 0$  on the principal branch. In particular,  $W'_0(0) = 1$  and  $W''_0(0) = -2$ .*

*Proof.* Write  $w = W_k(x)$ . Differentiating  $we^w = x$  gives

$$e^w(1 + w)w' = 1,$$

which proves the first expression in (20). Since  $e^{-w} = w/x$ , the second expression follows when  $xw \neq 0$ , and then extends by continuity wherever appropriate.

Now regard  $w'$  as a function of  $w$ :

$$w' = \frac{e^{-w}}{1 + w}.$$

Differentiate with respect to  $w$ , then multiply by  $w'$ :

$$w'' = \left( -\frac{e^{-w}}{1 + w} - \frac{e^{-w}}{(1 + w)^2} \right) \frac{e^{-w}}{1 + w} = -\frac{e^{-2w}(w + 2)}{(1 + w)^3}.$$

Substituting  $e^{-w} = w/x$  proves (22). The limits at zero follow either from these formulas or from the power series proved in [Section 8](#).  $\square$

**Corollary 5.2** (Monotonicity recovered from the derivative). *On  $(-1/e, \infty)$ ,  $W'_0(x) > 0$ . On  $(-1/e, 0)$ ,  $W'_{-1}(x) < 0$ . Moreover,*

$$\lim_{x \downarrow -1/e} W'_0(x) = +\infty, \quad \lim_{x \downarrow -1/e} W'_{-1}(x) = -\infty.$$

*Proof.* Use (20) and the ranges in [Theorem 2.1](#). At the common endpoint,  $1 + W_k(x) \rightarrow 0$ , with positive sign on the principal branch and negative sign on the nonprincipal branch. The remaining factors tend to nonzero finite values.  $\square$

**Theorem 5.3** (Concavity and the inflection point). *The principal branch  $W_0$  is strictly concave on  $(-1/e, \infty)$ . The branch  $W_{-1}$  is*

- *strictly convex on  $(-1/e, -2/e^2)$ ,*
- *has an inflection point at  $x = -2/e^2$ , where  $W_{-1}(x) = -2$ ,*
- *strictly concave on  $(-2/e^2, 0)$ .*

*Proof.* For  $W_0$ , one has  $w > -1$ , hence  $w + 2 > 0$  and  $(1 + w)^3 > 0$ . Formula (21) therefore gives  $w'' < 0$ .

For  $W_{-1}$ , one has  $w < -1$ , so  $(1 + w)^3 < 0$ . If  $-2 < w < -1$ , then  $w + 2 > 0$ , and (21) gives  $w'' > 0$ . If  $w < -2$ , then  $w + 2 < 0$ , and  $w'' < 0$ . The value  $w = -2$  corresponds to  $x = we^w = -2/e^2$ . The branch is decreasing, so the stated  $x$ -intervals follow.  $\square$

## 5.2 Higher derivatives

Repeated differentiation produces a structured family of polynomials.

**Theorem 5.4** (Polynomial recurrence for all derivatives). *Define polynomials  $P_n$  by*

$$P_1(w) = 1, \quad P_{n+1}(w) = (1+w)P'_n(w) - (nw + 3n - 1)P_n(w).$$

*Then, for every  $n \geq 1$ ,*

$$\frac{d^n}{dx^n} W_k(x) = \frac{e^{-nW_k(x)} P_n(W_k(x))}{(1 + W_k(x))^{2n-1}}. \quad (23)$$

*The first polynomials are*

$$\begin{aligned} P_1 &= 1, & P_2 &= -(w + 2), & P_3 &= 2w^2 + 8w + 9, \\ P_4 &= -(6w^3 + 36w^2 + 79w + 64). \end{aligned}$$

*Proof.* The formula for  $n = 1$  is (20). Assume (23) holds for  $n$ , and differentiate it first with respect to  $w$ :

$$\frac{d}{dw} \left( \frac{e^{-nw} P_n(w)}{(1+w)^{2n-1}} \right) = \frac{e^{-nw}}{(1+w)^{2n}} ((1+w)P'_n(w) - (nw + 3n - 1)P_n(w)).$$

Multiplication by  $dw/dx = e^{-w}/(1+w)$  yields

$$\frac{e^{-(n+1)w} P_{n+1}(w)}{(1+w)^{2n+1}},$$

which is (23) with  $n + 1$ . The displayed polynomials follow by direct substitution into the recurrence.  $\square$

## 5.3 Conditioning

Suppose  $x$  is perturbed by a small amount  $\Delta x$ . Linearization suggests

$$\Delta W \approx W'(x)\Delta x.$$

For nonzero  $x$  and  $W$ , the relative condition number is

$$\kappa_{\text{rel}}(x) = \left| \frac{xW'(x)}{W(x)} \right| = \frac{1}{|1 + W(x)|}. \quad (24)$$

*Proof.* Insert (20) into the definition of relative condition number.  $\square$

Thus both real branches are badly conditioned near  $-1/e$ , where  $W = -1$ . By contrast,  $\kappa_{\text{rel}} \rightarrow 0$  both as  $x \rightarrow +\infty$  on  $W_0$  and as  $x \rightarrow 0^-$  on  $W_{-1}$ . This distinction is important: the branch point is intrinsically sensitive, while the logarithmic singularity of  $W_{-1}$  at zero is comparatively well conditioned in a relative sense.

## 6 Integral calculus and exact moments

The substitution

$$w = W_k(x), \quad x = we^w, \quad dx = e^w(1+w)dw$$

turns many apparently non-elementary integrals into an exponential times a polynomial.

**Theorem 6.1** (Two basic antiderivatives). *On any interval contained in a differentiable branch,*

$$\int W_k(x) dx = e^{W_k(x)} (W_k(x)^2 - W_k(x) + 1) + C \quad (25)$$

$$= x \left( W_k(x) - 1 + \frac{1}{W_k(x)} \right) + C, \quad (26)$$

where the second form is used away from  $x = 0$ . Also, away from  $x = 0$ ,

$$\int \frac{W_k(x)}{x} dx = W_k(x) + \frac{1}{2} W_k(x)^2 + C. \quad (27)$$

*Proof.* With  $x = we^w$ ,

$$\int W_k(x) dx = \int we^w(1+w)dw = \int (w + w^2)e^w dw.$$

Direct differentiation shows that

$$\frac{d}{dw} [e^w(w^2 - w + 1)] = e^w(w + w^2),$$

which proves (25). Since  $e^w = x/w$ , it also proves (26). Furthermore,

$$\frac{dx}{x} = \frac{1+w}{w} dw,$$

and therefore

$$\int \frac{W_k(x)}{x} dx = \int w \frac{1+w}{w} dw = \int (1+w) dw = w + \frac{1}{2} w^2 + C.$$

□

**Corollary 6.2** (Two branch-dependent definite integrals). *The two real branches give*

$$\int_{-1/e}^0 W_0(x) dx = 1 - \frac{3}{e}, \quad (28)$$

$$\int_{-1/e}^0 W_{-1}(x) dx = -\frac{3}{e}. \quad (29)$$

*Both integrals are improper only in the harmless sense that  $W_{-1}(x) \rightarrow -\infty$  at 0; the second integral nevertheless converges.*

*Proof.* Use the antiderivative  $F(w) = e^w(w^2 - w + 1)$ . On the principal branch,  $w$  runs from  $-1$  to  $0$ , so

$$F(0) - F(-1) = 1 - \frac{3}{e}.$$

On the nonprincipal branch,  $w$  runs from  $-1$  to  $-\infty$ . Since  $e^w$  decreases faster than every polynomial grows,  $\lim_{w \rightarrow -\infty} F(w) = 0$ . Hence the integral is  $0 - 3/e$ . □

## 6.1 A finite formula for polynomial moments

For  $a \neq 0$  and an integer  $q \geq 0$ , define

$$Q_q^{(a)}(w) = \sum_{j=0}^q \frac{(-1)^j q!}{(q-j)! a^{j+1}} w^{q-j}. \quad (30)$$

**Lemma 6.3.** For  $a \neq 0$  and  $q \geq 0$ ,

$$\frac{d}{dw} \left( e^{aw} Q_q^{(a)}(w) \right) = e^{aw} w^q.$$

*Proof.* Differentiate the finite sum. The term  $aQ_q^{(a)}$  contributes

$$\sum_{j=0}^q \frac{(-1)^j q!}{(q-j)! a^j} w^{q-j}.$$

The derivative  $Q_q^{(a)'}$  contributes

$$\sum_{j=0}^{q-1} \frac{(-1)^j q!}{(q-j-1)! a^{j+1}} w^{q-j-1}.$$

Replace  $j$  by  $j-1$  in the second sum. Every term except  $w^q$  cancels with the corresponding term of the first sum.  $\square$

**Theorem 6.4** (All nonnegative integral moments). *Let  $m, n$  be nonnegative integers, put  $a = m+1$  and  $q = m+n$ . Then*

$$\int x^m W_k(x)^n dx = e^{aW_k(x)} \left( Q_q^{(a)}(W_k(x)) + Q_{q+1}^{(a)}(W_k(x)) \right) + C. \quad (31)$$

*Thus every integral of a nonnegative integer power of  $x$  times a nonnegative integer power of  $W_k(x)$  has an elementary expression in  $W_k(x)$ .*

*Proof.* With  $x = we^w$ ,

$$x^m W_k(x)^n dx = w^{m+n} e^{mw} e^w (1+w) dw = e^{aw} (w^q + w^{q+1}) dw.$$

Apply [Theorem 6.3](#) to the two terms.  $\square$

*Remark 6.5.* The same method works when the exponent of  $x$  is an integer less than  $-1$ , provided the corresponding powers of  $w$  can be integrated. The special case  $m = -1$  is simpler and leads to (27). Fractional powers require a consistent choice of the real or complex power and generally lead to incomplete gamma functions.

## 6.2 Mellin-type definite integrals on the unbounded ends

The same substitution also evaluates a continuous family of improper integrals. Recall

$$\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt \quad (s > 0),$$

and define the upper incomplete gamma integral by

$$\Gamma(s, c) = \int_c^\infty t^{s-1} e^{-t} dt \quad (c > 0).$$

Because the latter integral starts at a positive number, it converges for every real  $s$ .

**Theorem 6.6** (Mellin integrals for both unbounded real ends). *Let  $\alpha, \beta \in \mathbb{R}$ .*

(i) *If  $\beta > 1$  and  $\alpha - \beta > -1$ , then*

$$\int_0^\infty \frac{W_0(x)^\alpha}{x^\beta} dx = \frac{\Gamma(\alpha - \beta + 1)}{(\beta - 1)^{\alpha - \beta + 1}} + \frac{\Gamma(\alpha - \beta + 2)}{(\beta - 1)^{\alpha - \beta + 2}} \quad (32)$$

$$= \frac{\alpha \Gamma(\alpha - \beta + 1)}{(\beta - 1)^{\alpha - \beta + 2}}. \quad (33)$$

(ii) *If  $\beta > -1$ , then*

$$\begin{aligned} \int_{-1/e}^0 [-W_{-1}(x)]^\alpha (-x)^\beta dx \\ = \frac{\Gamma(\alpha + \beta + 2, \beta + 1)}{(\beta + 1)^{\alpha + \beta + 2}} - \frac{\Gamma(\alpha + \beta + 1, \beta + 1)}{(\beta + 1)^{\alpha + \beta + 1}}. \end{aligned} \quad (34)$$

*Equivalently, using the incomplete-gamma recurrence,*

$$\int_{-1/e}^0 [-W_{-1}(x)]^\alpha (-x)^\beta dx = \frac{\alpha \Gamma(\alpha + \beta + 1, \beta + 1)}{(\beta + 1)^{\alpha + \beta + 2}} + \frac{e^{-(\beta + 1)}}{\beta + 1}. \quad (35)$$

*Proof.* For part (i), put  $w = W_0(x)$ . As  $x$  runs from 0 to  $\infty$ ,  $w$  runs from 0 to  $\infty$ , and

$$x = we^w, \quad dx = e^w(1 + w) dw.$$

Consequently, with  $c = \beta - 1 > 0$  and  $\nu = \alpha - \beta > -1$ ,

$$\begin{aligned} \int_0^\infty \frac{W_0(x)^\alpha}{x^\beta} dx &= \int_0^\infty w^\nu (1 + w) e^{-cw} dw \\ &= \frac{\Gamma(\nu + 1)}{c^{\nu + 1}} + \frac{\Gamma(\nu + 2)}{c^{\nu + 2}}. \end{aligned}$$

This proves (32). Since  $\Gamma(\nu + 2) = (\nu + 1)\Gamma(\nu + 1)$ , the two terms combine to (33). The hypotheses are also exactly what is needed for convergence at  $w = 0$  and at  $w = \infty$ .

For part (ii), put  $y = -W_{-1}(x)$ . Then  $y$  runs from 1 to  $\infty$ ,

$$x = -ye^{-y}, \quad dx = (y - 1)e^{-y} dy.$$

Writing  $p = \alpha + \beta$  and  $c = \beta + 1 > 0$ , we obtain

$$\begin{aligned} \int_{-1/e}^0 [-W_{-1}(x)]^\alpha (-x)^\beta dx &= \int_1^\infty y^p (y - 1) e^{-cy} dy \\ &= \int_1^\infty (y^{p+1} - y^p) e^{-cy} dy. \end{aligned}$$

In each term let  $t = cy$ . This gives (34). Finally, integration by parts proves

$$\Gamma(s + 1, c) = s\Gamma(s, c) + c^s e^{-c},$$

and substitution with  $s = \alpha + \beta + 1$  gives (35). □

**Example 6.7.** *Taking  $(\alpha, \beta) = (1, 3/2)$  in (33) gives the attractive identity*

$$\int_0^\infty \frac{W_0(x)}{x\sqrt{x}} dx = 2\sqrt{2\pi}. \quad (36)$$

*It is an exact evaluation of an integral whose integrand contains a non-elementary inverse function.*

## 7 Taylor series at an arbitrary regular point

The derivative recurrence of [Theorem 5.4](#) immediately gives a local series about any point away from the branch point.

**Theorem 7.1** (Local Taylor expansion). *Let  $x_0$  be a point on a branch  $W_k$ , and let  $w_0 = W_k(x_0) \neq -1$ . In a sufficiently small neighborhood of  $x_0$ ,*

$$W_k(x) = w_0 + \sum_{n=1}^{\infty} \frac{e^{-nw_0} P_n(w_0)}{n!(1+w_0)^{2n-1}} (x-x_0)^n, \quad (37)$$

where the polynomials  $P_n$  are those of [Theorem 5.4](#). This formula applies without change to  $W_0$  and  $W_{-1}$ , provided  $x_0$  belongs to the selected branch.

*Proof.* The inverse function theorem gives a differentiable, in fact analytic, local inverse of  $we^w$  at  $w_0$ , because  $e^{w_0}(1+w_0) \neq 0$ . Taylor's formula therefore applies, and [Theorem 5.4](#) supplies its coefficients.  $\square$

**Example 7.2.** At  $x_0 = 1$ ,  $w_0 = \Omega$ . The first two terms are

$$W_0(1+h) = \Omega + \frac{e^{-\Omega}}{1+\Omega} h - \frac{e^{-2\Omega}(\Omega+2)}{2(1+\Omega)^3} h^2 + \mathcal{O}(h^3).$$

Since  $e^{-\Omega} = \Omega$ , every coefficient can be written as a rational function of  $\Omega$ .

The singular point  $w_0 = -1$  is deliberately excluded. There the inverse has a square-root, not an ordinary Taylor, expansion; this is the subject of [Section 9](#).

## 8 The Maclaurin series of the principal branch

The principal branch is analytic at the origin. Its coefficients have an unexpectedly simple closed form.

For a formal power series  $A(t)$ , the notation  $[t^n]A(t)$  means the coefficient of  $t^n$ . We use the following form of Lagrange inversion, proved in [Section A](#).

**Theorem 8.1** (Lagrange–Buermann inversion). *Suppose  $y = x\phi(y)$ , where  $\phi(0) \neq 0$ , and let  $H$  be a formal power series. For  $n \geq 1$ ,*

$$[x^n]H(y(x)) = \frac{1}{n} [t^{n-1}]H'(t)\phi(t)^n. \quad (38)$$

In particular, for integers  $m \geq 1$  and  $n \geq m$ ,

$$[x^n]y(x)^m = \frac{m}{n} [t^{n-m}]\phi(t)^n. \quad (39)$$

The defining equation for  $W$  can be written

$$W = xe^{-W}.$$

Thus  $\phi(t) = e^{-t}$ .

**Theorem 8.2** (Maclaurin series and powers). *For  $|x| < 1/e$ ,*

$$\boxed{W_0(x) = \sum_{n=1}^{\infty} \frac{(-n)^{n-1}}{n!} x^n} = x - x^2 + \frac{3}{2}x^3 - \frac{8}{3}x^4 + \frac{125}{24}x^5 - \dots \quad (40)$$

*More generally, for every integer  $m \geq 1$ ,*

$$W_0(x)^m = \sum_{n=m}^{\infty} \frac{m}{n} \frac{(-n)^{n-m}}{(n-m)!} x^n. \quad (41)$$

*Consequently*

$$W_0^{(n)}(0) = (-n)^{n-1} \quad (n \geq 1). \quad (42)$$

*The radius of convergence is exactly  $1/e$ .*

*Proof.* Apply (39) with  $\phi(t) = e^{-t}$ :

$$[x^n]W(x)^m = \frac{m}{n} [t^{n-m}]e^{-nt} = \frac{m}{n} \frac{(-n)^{n-m}}{(n-m)!}.$$

This proves (41), and  $m = 1$  gives (40). The derivative formula follows by multiplying the coefficient of  $x^n$  by  $n!$ .

Let

$$a_n = \frac{n^{n-1}}{n!}.$$

Then

$$\frac{a_{n+1}}{a_n} = \left(1 + \frac{1}{n}\right)^{n-1} \rightarrow e.$$

The ratio test therefore gives radius  $1/e$ . Within that disk, the series produced by Lagrange inversion satisfies  $W = xe^{-W}$  and has  $W(0) = 0$ . The local inverse is unique, so it is the principal branch.  $\square$

*Remark 8.3* (What the radius means). The nearest singularity to the origin is the branch point  $-1/e$ . The coefficient calculation detects exactly that distance without requiring us to know the complex geometry in advance.

## 8.1 Rigorous remainder estimates

Let

$$T_n(x) = \frac{(-n)^{n-1}}{n!} x^n.$$

A direct calculation gives

$$\left| \frac{T_{n+1}(x)}{T_n(x)} \right| = |x| \left(1 + \frac{1}{n}\right)^{n-1} < e|x|. \quad (43)$$

**Theorem 8.4** (Geometric tail bound). *If  $|x| < 1/e$ ,  $q = e|x| < 1$ , and  $S_N(x) = \sum_{n=1}^N T_n(x)$ , then*

$$|W_0(x) - S_N(x)| \leq \frac{|T_{N+1}(x)|}{1-q}. \quad (44)$$

*If  $0 \leq x \leq 1/e$ , the terms alternate with decreasing magnitudes, so the sharper estimate*

$$|W_0(x) - S_N(x)| \leq |T_{N+1}(x)| \quad (45)$$

*holds, and the error has the sign of  $T_{N+1}(x)$ .*



*Proof.* Equation (43) shows that every successive magnitude in the tail is at most  $q$  times the preceding magnitude. Summing the resulting geometric majorant proves (44). For  $x \geq 0$ , the factor  $(-n)^{n-1}$  makes the signs alternate. The same ratio is at most 1 on  $0 \leq x \leq 1/e$ , so the alternating-series theorem proves (45).  $\square$

For negative  $x$ , every nonzero term in (40) is negative. At the endpoint  $x = -1/e$ , the series still converges absolutely. One way to see this without invoking the full Stirling formula is Raabe's test: if

$$b_n = \frac{n^{n-1}}{n!e^n},$$

then

$$\log \frac{b_n}{b_{n+1}} = 1 - (n-1) \log \left(1 + \frac{1}{n}\right) = \frac{3}{2n} + \mathcal{O}(n^{-2}).$$

Hence  $b_n/b_{n+1} = 1 + 3/(2n) + \mathcal{O}(n^{-2})$ , and Raabe's test applies because  $3/2 > 1$ . Moreover, for every  $x \in [-1/e, 0]$ , the magnitude of the  $n$ -th term is at most  $b_n$ . The Weierstrass  $M$ -test therefore gives uniform convergence on this closed interval. The sum is continuous there, agrees with  $W_0$  on  $(-1/e, 0]$ , and hence agrees at the left endpoint by continuity. Consequently

$$\sum_{n=1}^{\infty} \frac{(-n)^{n-1}}{n!} \left(-\frac{1}{e}\right)^n = W_0(-1/e) = -1.$$

## 8.2 The tree function and Cayley's formula

Define

$$T(x) := -W_0(-x). \tag{46}$$

Then

$$T(x) = \sum_{n=1}^{\infty} \frac{n^{n-1}}{n!} x^n, \quad T = xe^T.$$

The coefficient  $n^{n-1}$  counts rooted labelled trees on  $n$  vertices.

**Theorem 8.5** (Cayley and the tree coefficients). *There are  $n^{n-2}$  unrooted trees on the labelled vertex set  $\{1, \dots, n\}$ , and  $n^{n-1}$  rooted labelled trees. Therefore the exponential generating function for rooted labelled trees is exactly  $T(x) = -W_0(-x)$ .*

*Proof.* The Prufer code of a labelled tree is formed by repeatedly deleting the smallest-labelled leaf and recording its neighbor, until two vertices remain. This produces a sequence of length  $n-2$  with entries in  $\{1, \dots, n\}$ .

Conversely, given such a sequence, repeatedly join its first entry to the smallest label not appearing in the remaining sequence, delete that first entry, and finally join the last two unused labels. These two procedures are inverse to each other. Thus labelled trees are in bijection with the  $n^{n-2}$  sequences of length  $n-2$ . Choosing one of the  $n$  vertices as root gives  $n \cdot n^{n-2} = n^{n-1}$  rooted trees.

It remains only to compare coefficients: (46) and (40) give  $n^{n-1}/n!$ , the exponential-generating-function coefficient for  $n^{n-1}$  labelled objects.  $\square$

There is also a structural explanation of  $T = xe^T$ . A rooted labelled tree consists of a root together with an unordered set of rooted labelled trees attached to it. In the language of exponential generating functions, an unordered labelled set contributes an exponential, giving  $T = x \exp(T)$ . Lagrange inversion then recovers Cayley's coefficient.

## 9 The square-root expansion at the branch point

Ordinary Taylor series fail at  $x = -1/e$  because the derivative of the inverse is infinite there. The correct local variable is a square root.

Put

$$w = -1 + s.$$

Then

$$ex = (-1 + s)e^s,$$

and expansion at  $s = 0$  gives

$$2(1 + ex) = \sum_{m=2}^{\infty} \frac{2(m-1)}{m!} s^m = s^2 + \frac{2}{3}s^3 + \frac{1}{4}s^4 + \frac{1}{15}s^5 + \dots \quad (47)$$

Define the *signed* square-root coordinates

$$p_0(x) = +\sqrt{2(1 + ex)}, \quad p_{-1}(x) = -\sqrt{2(1 + ex)}. \quad (48)$$

**Theorem 9.1** (Unified Puiseux expansion). *There is a convergent power series  $S(p) = \sum_{n \geq 1} a_n p^n$  near  $p = 0$  such that*

$$W_k(x) = -1 + S(p_k(x)), \quad k \in \{0, -1\}. \quad (49)$$

*Its first terms are*

$$S(p) = p - \frac{1}{3}p^2 + \frac{11}{72}p^3 - \frac{43}{540}p^4 + \frac{769}{17280}p^5 - \frac{221}{8505}p^6 + \mathcal{O}(p^7). \quad (50)$$

*Thus the same series describes both real branches: only the sign of  $p$  changes.*

*Proof.* For  $s \neq 0$ , divide (47) by  $s^2$  and choose the analytic square root that equals 1 at  $s = 0$ :

$$p = s \sqrt{1 + \frac{2}{3}s + \frac{1}{4}s^2 + \dots}$$

This defines an analytic function  $p = p(s)$  with  $p(0) = 0$  and  $p'(0) = 1$ . The inverse function theorem gives an analytic inverse  $s = S(p)$  near 0, with  $S(0) = 0$  and  $S'(0) = 1$ . For the principal branch  $s \geq 0$ , so  $p = p_0 \geq 0$ . For the nonprincipal branch  $s \leq 0$ , so the analytic signed coordinate is  $p = p_{-1} \leq 0$ . Hence (49) holds.

To compute coefficients, substitute  $S(p) = a_1 p + a_2 p^2 + \dots$  into (47) and compare successive powers of  $p$ . This gives  $a_1 = 1$ , then

$$a_2 = -\frac{1}{3}, \quad a_3 = \frac{11}{72}, \quad a_4 = -\frac{43}{540}, \quad a_5 = \frac{769}{17280}, \quad a_6 = -\frac{221}{8505}.$$

Because each new comparison is linear in the next unknown coefficient, the procedure determines all coefficients uniquely.  $\square$

Here is an explicit recurrence that makes the final sentence of the proof algorithmic. Write

$$S(p) = \sum_{n \geq 1} a_n p^n, \quad e^{-S(p)} = \sum_{n \geq 0} b_n p^n.$$

Differentiating the defining relation gives

$$S(p)S'(p) = p e^{-S(p)}.$$

Consequently

$$b_0 = 1, \quad b_n = -\frac{1}{n} \sum_{j=1}^n j a_j b_{n-j}, \quad (51)$$

$$a_1 = 1, \quad a_n = \frac{b_{n-1} - \sum_{j=2}^{n-1} (n-j+1) a_j a_{n-j+1}}{n+1} \quad (n \geq 2). \quad (52)$$

*Proof.* The identity  $B' = -S'B$ , where  $B = e^{-S}$ , gives (51) by comparing the coefficient of  $p^{n-1}$ . The coefficient of  $p^n$  in  $SS' = pB$  is

$$\sum_{j=1}^n (n-j+1) a_j a_{n-j+1} = b_{n-1}.$$

The terms  $j = 1$  and  $j = n$  together equal  $(n+1)a_n$ ; moving the remaining terms to the other side gives (52).  $\square$

## 9.1 Even and odd combinations of the two branches

Let

$$q = \sqrt{2(1+ex)} \geq 0.$$

Then  $p_0 = q$  and  $p_{-1} = -q$ . Therefore

$$W_0(x) - W_{-1}(x) = 2q + \frac{11}{36}q^3 + \frac{769}{8640}q^5 + \mathcal{O}(q^7), \quad (53)$$

$$\frac{W_0(x) + W_{-1}(x)}{2} = -1 - \frac{1}{3}q^2 - \frac{43}{540}q^4 - \frac{221}{8505}q^6 + \mathcal{O}(q^8). \quad (54)$$

The branch difference contains only odd powers, while the branch mean contains only even powers. These formulas are immediate by adding and subtracting  $S(q)$  and  $S(-q)$ .

The leading term gives the especially useful equivalence

$$W_{0,-1}(x) = -1 \pm \sqrt{2e \left( x + \frac{1}{e} \right)} + \mathcal{O} \left( x + \frac{1}{e} \right), \quad (55)$$

where the plus sign is for  $W_0$ . It also explains the vertical tangents:

$$W'_0(x) \sim \frac{e}{q}, \quad W'_{-1}(x) \sim -\frac{e}{q}.$$

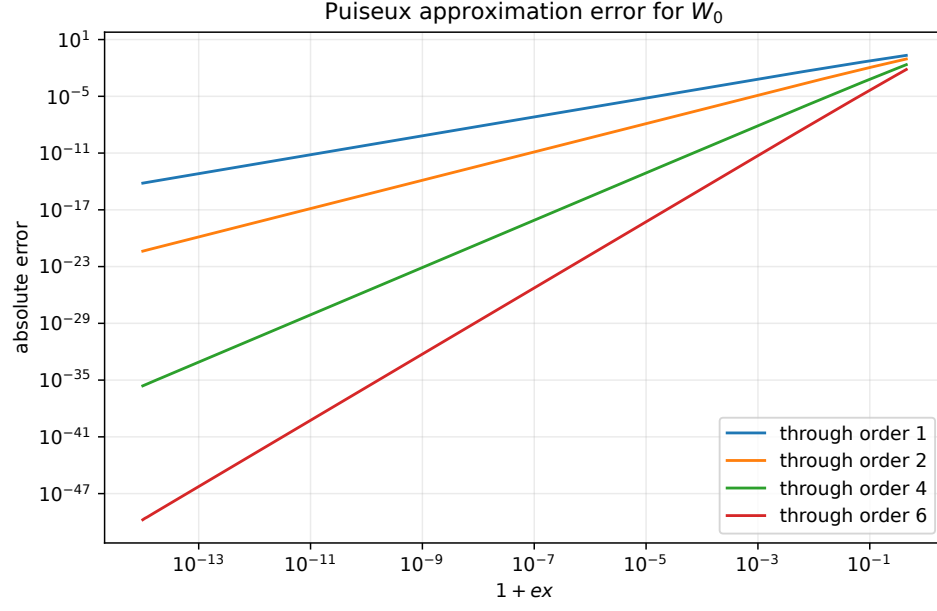


Figure 2: Absolute errors of successive truncations of (50) on the principal branch.

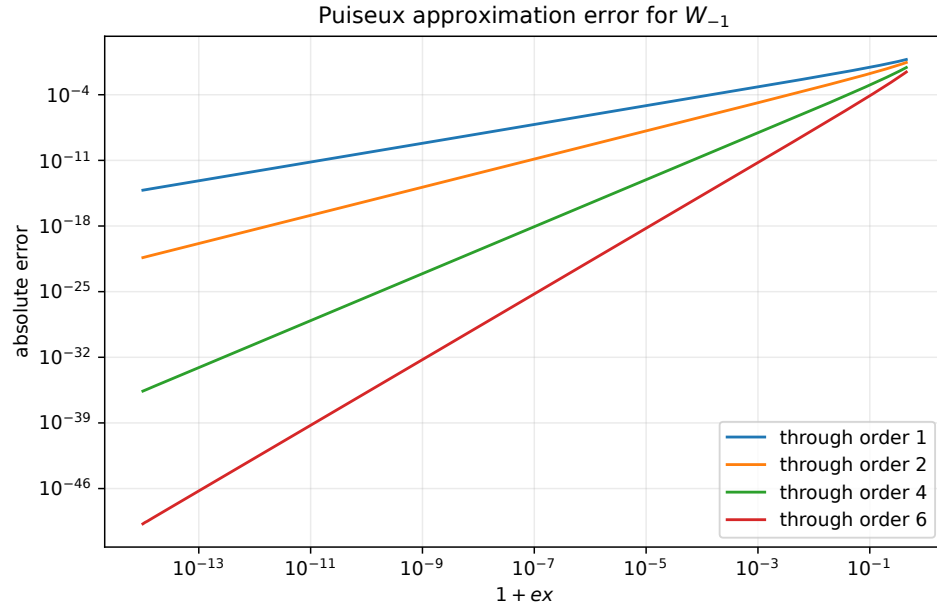


Figure 3: The same signed expansion, now with  $p < 0$ , approximates the nonprincipal branch.

## 9.2 An exponential branch-point coordinate

A second useful parametrization is

$$x = -e^{-1-t^2/2}. \quad (56)$$

Choose  $t > 0$  for  $W_0$  and  $t < 0$  for  $W_{-1}$ . If  $W = -1 + u$ , then the defining equation becomes

$$(1 - u)e^u = e^{-t^2/2},$$

or

$$\frac{t^2}{2} = -u - \log(1 - u). \quad (57)$$

The right side begins with  $u^2/2$ , so another inverse-series calculation gives

$$W = -1 + t - \frac{1}{3}t^2 + \frac{1}{36}t^3 + \frac{1}{270}t^4 + \frac{1}{4320}t^5 - \frac{1}{17010}t^6 + \mathcal{O}(t^7). \quad (58)$$

The proof is identical in structure to that of [Theorem 9.1](#): the signed analytic square root of  $2[-u - \log(1 - u)]$  has derivative 1 at the origin and therefore has a local analytic inverse.

*Literature note 9.2.* The NIST DLMF [7] and the foundational paper of Corless et al. [5] tabulate many more branch-point coefficients. The exponential coordinate (56) is particularly convenient in rigorous numerical work, because  $t^2/2 = -1 - \log(-x)$  is computed accurately even when  $x$  is represented in logarithmic form.

## 10 Logarithmic asymptotic expansions

The two unbounded real limits are

$$W_0(x) \longrightarrow +\infty \quad (x \rightarrow +\infty), \quad W_{-1}(x) \longrightarrow -\infty \quad (x \rightarrow 0^-).$$

They are governed by the same algebra. The first approximation follows by taking logarithms:

$$W + \log |W| = \log |x|.$$

If  $|W|$  is large, then  $W$  is approximately  $\log |x|$ , and a second approximation subtracts the logarithm of that first approximation.

### 10.1 Unsigned Stirling numbers

The unsigned Stirling number of the first kind  $\left[ \begin{smallmatrix} n \\ j \end{smallmatrix} \right]$  is defined by

$$t(t+1) \cdots (t+n-1) = \sum_{j=0}^n \left[ \begin{smallmatrix} n \\ j \end{smallmatrix} \right] t^j. \quad (59)$$

It counts permutations of  $n$  objects having  $j$  cycles, but only its algebraic role is needed here. The recurrence

$$\left[ \begin{smallmatrix} n \\ j \end{smallmatrix} \right] = \left[ \begin{smallmatrix} n-1 \\ j-1 \end{smallmatrix} \right] + (n-1) \left[ \begin{smallmatrix} n-1 \\ j \end{smallmatrix} \right] \quad (60)$$

follows by multiplying the product for  $n-1$  by  $t+n-1$ .

**Lemma 10.1** (Exponential generating function). *For  $r \geq 0$ ,*

$$\frac{[-\log(1-z)]^r}{r!} = \sum_{n=r}^{\infty} \left[ \begin{smallmatrix} n \\ r \end{smallmatrix} \right] \frac{z^n}{n!}, \quad |z| < 1. \quad (61)$$

*Consequently,*

$$\frac{[-\log(1-z)]^r}{r!(1-z)} = \sum_{n=r}^{\infty} \left[ \begin{smallmatrix} n+1 \\ r+1 \end{smallmatrix} \right] \frac{z^n}{n!}. \quad (62)$$

*Proof.* Let the right side of (61) be  $F_r(z)$ . Using (60) and shifting indices shows

$$F'_r(z) = \frac{F_{r-1}(z)}{1-z}, \quad F_r(0) = 0 \quad (r \geq 1),$$

while  $F_0(z) = 1$ . The functions  $[-\log(1-z)]^r/r!$  satisfy the same differential recurrence and initial conditions, so induction on  $r$  proves (61). Differentiate the formula with  $r+1$  in place of  $r$  to obtain (62).  $\square$

Define polynomials

$$\mathcal{P}_n(L) = \sum_{m=1}^n \frac{(-1)^{n-m}}{m!} \left[ \begin{matrix} n \\ n-m+1 \end{matrix} \right] L^m. \quad (63)$$

The first four are

$$\mathcal{P}_1(L) = L, \quad (64)$$

$$\mathcal{P}_2(L) = \frac{L(-2+L)}{2}, \quad (65)$$

$$\mathcal{P}_3(L) = \frac{L(6-9L+2L^2)}{6}, \quad (66)$$

$$\mathcal{P}_4(L) = \frac{L(-12+36L-22L^2+3L^3)}{12}. \quad (67)$$

**Theorem 10.2** (Unified complete real asymptotic expansion). *For either of the following limits,*

| <i>branch and limit</i>         | $L_1$      | $L_2$             |
|---------------------------------|------------|-------------------|
| $W_0(x), x \rightarrow +\infty$ | $\log x$   | $\log \log x$     |
| $W_{-1}(x), x \rightarrow 0^-$  | $\log(-x)$ | $\log[-\log(-x)]$ |

one has, for every fixed  $N \geq 1$ ,

$$W(x) = L_1 - L_2 + \sum_{n=1}^N \frac{\mathcal{P}_n(L_2)}{L_1^n} + \mathcal{O}\left(\frac{L_2^{N+1}}{|L_1|^{N+1}}\right). \quad (68)$$

Equivalently, the formal complete expansion is

$$W(x) \sim L_1 - L_2 + \sum_{k=0}^{\infty} \sum_{m=1}^{\infty} \frac{(-1)^k}{m!} \left[ \begin{matrix} k+m \\ k+1 \end{matrix} \right] \frac{L_2^m}{L_1^{k+m}}. \quad (69)$$

*Proof.* Set

$$W = L_1 - L_2 + v, \quad \sigma = \frac{1}{L_1}, \quad \tau = \frac{L_2}{L_1}.$$

For the principal positive branch,  $\log x = W + \log W$ . Factoring  $L_1$  from  $W$  gives

$$0 = v + \log(1 - \tau + \sigma v).$$

For the negative branch,  $\log(-x) = W + \log(-W)$ ; because  $L_1 < 0$ , factoring  $-L_1$  from  $-W$  gives exactly the same equation. Hence

$$\tau = 1 - e^{-v} + \sigma v. \quad (70)$$

Apply Lagrange inversion to the inverse of the right side:

$$v = \sum_{m=1}^{\infty} \frac{\tau^m}{m} [t^{m-1}] \left( \frac{t}{1 - e^{-t} + \sigma t} \right)^m$$

$$= \sum_{m=1}^{\infty} \sum_{k=0}^{\infty} \frac{(-1)^k}{m} \binom{m+k-1}{k} \sigma^k \tau^m [t^{m-1}] \left( \frac{t}{1-e^{-t}} \right)^{m+k}. \quad (71)$$

We claim that

$$[t^{m-1}] \left( \frac{t}{1-e^{-t}} \right)^{m+k} = \frac{k!}{(m+k-1)!} \begin{bmatrix} m+k \\ k+1 \end{bmatrix}. \quad (72)$$

To prove it, write the coefficient as a formal residue:

$$\text{Res}_{t=0} \frac{t^k}{(1-e^{-t})^{m+k}} dt.$$

Make the substitution  $z = 1 - e^{-t}$ , so  $t = -\log(1-z)$  and  $dt = dz/(1-z)$ . The residue becomes the coefficient of  $z^{m+k-1}$  in

$$\frac{[-\log(1-z)]^k}{1-z}.$$

Formula (62) says that this coefficient is the right side of (72).

Substitution into (71) simplifies the numerical factor:

$$\frac{1}{m} \binom{m+k-1}{k} \frac{k!}{(m+k-1)!} = \frac{1}{m!}.$$

Therefore

$$v = \sum_{m \geq 1} \sum_{k \geq 0} \frac{(-1)^k}{m!} \begin{bmatrix} m+k \\ k+1 \end{bmatrix} \sigma^k \tau^m.$$

Since  $\sigma^k \tau^m = L_2^m / L_1^{k+m}$ , this is (69); collecting terms with  $k+m=n$  gives (63).

Finally, the implicit function in (70) has derivative 1 with respect to  $v$  at  $(v, \sigma, \tau) = (0, 0, 0)$ . Hence  $v$  is analytic in  $(\sigma, \tau)$  near the origin. Its Taylor remainder after total degree  $N$  is  $\mathcal{O}((|\sigma| + |\tau|)^{N+1})$ . In both limits,  $\sigma \rightarrow 0$ ,  $\tau \rightarrow 0$ , and  $|\tau| = L_2/|L_1|$  eventually dominates  $|\sigma|$ . This proves the remainder estimate in (68).  $\square$

## 10.2 The expansion written separately for each branch

For  $x \rightarrow +\infty$ , set  $A = \log x$  and  $B = \log A$ . Then

$$\begin{aligned} W_0(x) = & A - B + \frac{B}{A} + \frac{B(-2+B)}{2A^2} + \frac{B(6-9B+2B^2)}{6A^3} \\ & + \frac{B(-12+36B-22B^2+3B^3)}{12A^4} + \mathcal{O}\left(\frac{B^5}{A^5}\right). \end{aligned} \quad (73)$$

For  $x \rightarrow 0^-$ , set

$$\eta = \log \frac{1}{-x}, \quad B = \log \eta.$$

Since  $L_1 = -\eta$ ,

$$\begin{aligned} W_{-1}(x) = & -\eta - B - \frac{B}{\eta} + \frac{B(B-2)}{2\eta^2} - \frac{B(2B^2-9B+6)}{6\eta^3} \\ & + \frac{B(3B^3-22B^2+36B-12)}{12\eta^4} + \mathcal{O}\left(\frac{B^5}{\eta^5}\right). \end{aligned} \quad (74)$$

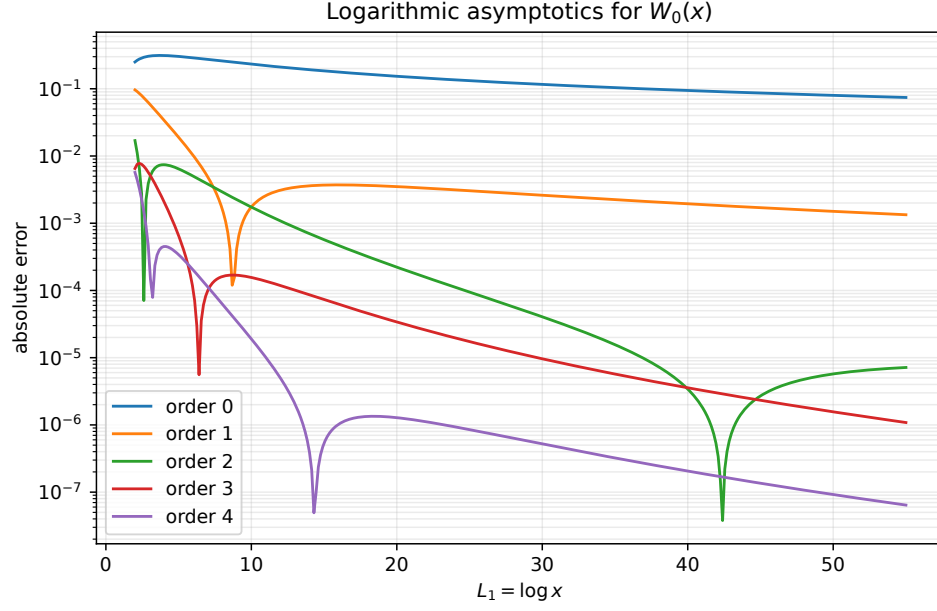


Figure 4: Truncation errors in the logarithmic expansion for  $W_0$ . The horizontal coordinate is  $A = \log x$ , so the right edge represents astronomically large  $x$ .

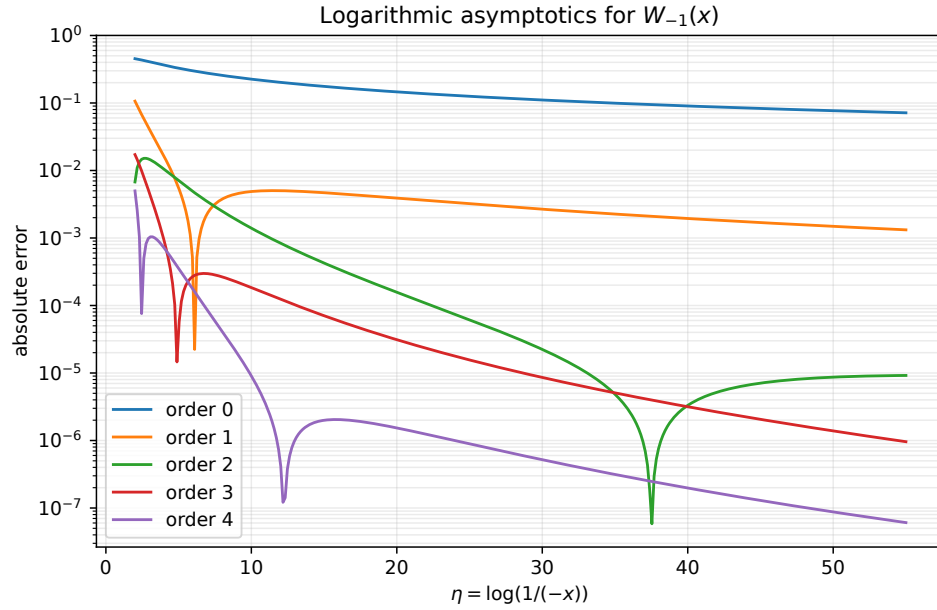


Figure 5: The same coefficient polynomials approximate  $W_{-1}(x)$  as  $x \rightarrow 0^-$ , after the sign of  $L_1$  is changed.

*Remark 10.3* (Asymptotic does not mean “keep every term”). For a fixed moderate argument, adding terms can eventually stop improving an asymptotic approximation. One should compare successive terms and stop near the smallest one. For genuinely large  $|L_1|$ , however, each fixed-order truncation in [Theorem 10.2](#) has the proved error order shown there.



## 11 Rigorous elementary bounds

Bounds serve three purposes: they explain the shape of the graph, provide quick approximations, and furnish guaranteed starting values for iteration.

**Lemma 11.1** (Two logarithm inequalities). *For  $t > -1$ ,*

$$\frac{t}{1+t} \leq \log(1+t) \leq t, \quad (75)$$

*with equality only at  $t = 0$ .*

*Proof.* For the upper bound, let  $g(t) = t - \log(1+t)$ . Then  $g'(t) = t/(1+t)$ , so  $g$  decreases to  $g(0) = 0$  on  $(-1, 0)$  and increases from 0 on  $(0, \infty)$ . Thus  $g \geq 0$ .

For the lower bound, let  $h(t) = \log(1+t) - t/(1+t)$ . Then  $h'(t) = t/(1+t)^2$ . The same monotonicity argument gives  $h \geq 0$ .  $\square$

### 11.1 Bounds for the principal branch on $x \geq 0$

**Theorem 11.2** (Global elementary principal-branch bounds). *For every  $x \geq 0$ ,*

$$\boxed{\frac{x}{1+x} \leq W_0(x) \leq \log(1+x) \leq x.} \quad (76)$$

*All inequalities are strict when  $x > 0$ , except where an accidental equality is forced at  $x = 0$ .*

*Proof.* Let  $w = W_0(x) \geq 0$ , so  $x = we^w$ . The elementary inequality  $1 - e^{-w} \leq w$  is equivalent both to

$$\frac{x}{1+x} \leq w$$

and to

$$e^w \leq 1+x,$$

the latter being  $w \leq \log(1+x)$ . Finally  $\log(1+x) \leq x$  is [Theorem 11.1](#).  $\square$

The first rational lower bound is also the  $[1/1]$  Padé approximant of  $W_0$  at the origin. Its role is therefore both analytic and numerical.

**Theorem 11.3** (Logarithmic bounds for large positive arguments). *Let  $x > e$ ,  $L = \log x > 1$ . Then*

$$L - \log L < W_0(x) < L - \log(L - \log L) < L. \quad (77)$$

*At  $x = e$ , the first lower bound is an equality:  $W_0(e) = 1$ .*

*Proof.* The function  $f(w) = we^w$  is increasing for  $w > 0$ . Since

$$Le^L = Lx > x,$$

we have  $W_0(x) < L$ . Put  $a = L - \log L > 0$ . Then

$$ae^a = (L - \log L) \frac{x}{L} = x \left(1 - \frac{\log L}{L}\right) < x,$$

so  $a < W_0(x)$ . The equation  $W_0 = L - \log W_0$  and the lower bound  $W_0 > a$  imply

$$W_0 = L - \log W_0 < L - \log a.$$

Finally, the function  $h(L) = L - \log L$  is strictly increasing for  $L > 1$ , and  $h(1) = 1$ . Thus  $a = h(L) > 1$ , so  $\log a > 0$  and  $L - \log a < L$ .  $\square$

Repeated use of the decreasing map  $g(w) = L - \log w$  produces alternating upper and lower bounds. For example, starting with

$$\ell_0 = L - \log L, \quad u_0 = L,$$

the implication

$$\ell \leq W_0 \leq u \implies L - \log u \leq W_0 \leq L - \log \ell$$

follows immediately from the fact that  $g$  is decreasing.

## 11.2 Bounds on the negative part of the principal branch

**Theorem 11.4** (Two square-root bounds and a linear bound). *For  $-1/e < x < 0$ ,*

$$-1 + \sqrt{1 + ex} < W_0(x) < -1 + \sqrt{2(1 + ex)}, \quad (78)$$

and also

$$W_0(x) < x. \quad (79)$$

*The first lower bound becomes exact at both endpoints in the limiting sense.*

*Proof.* Write  $w = -1 + s$  with  $0 < s < 1$ . Then

$$1 + ex = 1 + (s - 1)e^s.$$

The lower inequality is equivalent to

$$s^2 > 1 + (s - 1)e^s.$$

Since  $s - 1 < 0$ , this in turn is equivalent to  $1 + s < e^s$ , which is strict for  $s > 0$ .

For the upper square-root bound, define

$$H(s) = 2[1 - (1 - s)e^s] - s^2.$$

Then  $H(0) = 0$  and

$$H'(s) = 2s(e^s - 1) > 0$$

for  $s > 0$ . Thus  $s^2 < 2(1 + ex)$ , proving the upper bound.

Finally  $x = we^w$ . Since  $w < 0$  and  $0 < e^w < 1$ , multiplication of the negative number  $w$  by  $e^w$  moves it toward zero, so  $w < x$ .  $\square$

Near  $x = 0$ , the bound  $W_0(x) < x$  is much sharper than the upper square-root bound; near the branch point, the square-root bounds reveal the correct scale.

## 11.3 Bounds for the nonprincipal branch near the branch point

The following result, due to Chatzigeorgiou [3], is particularly useful because its proof needs only elementary inequalities.

**Theorem 11.5** (Two-sided  $W_{-1}$  bound). *For  $u > 0$ ,*

$$\boxed{-1 - \sqrt{2u} - u < W_{-1}(-e^{-1-u}) < -1 - \sqrt{2u} - \frac{2}{3}u.} \quad (80)$$

*Proof.* Write

$$W_{-1}(-e^{-1-u}) = -1 - v, \quad v > 0.$$

The defining equation gives

$$(1 + v)e^{-v} = e^{-u},$$

hence

$$u = v - \log(1 + v). \quad (81)$$

The desired inequalities are equivalent to

$$\sqrt{2u} + \frac{2}{3}u < v < \sqrt{2u} + u. \quad (82)$$

For the upper bound,  $v - u = \log(1 + v)$ . Put  $t = \log(1 + v) > 0$ , so  $v = e^t - 1$  and  $u = e^t - 1 - t$ . The strict exponential inequality  $e^t > 1 + t + t^2/2$  gives

$$t < \sqrt{2u},$$

and hence  $v = u + t < u + \sqrt{2u}$ .

For the lower bound, note first that  $v - \frac{2}{3}u = (v + 2t)/3 > 0$ . It is therefore enough to square. The required inequality is equivalent to

$$(v + 2t)^2 > 18u.$$

Substituting  $v = e^t - 1$  and  $u = e^t - 1 - t$ , define

$$F(t) = (e^t - 1 + 2t)^2 - 18(e^t - 1 - t).$$

A direct differentiation gives

$$F(0) = F'(0) = F''(0) = F'''(0) = 0,$$

and

$$F^{(4)}(t) = 4e^t(t + 4e^t - 1) > 0 \quad (t > 0).$$

Successive integrations from 0 show  $F'''(t) > 0$ , then  $F''(t) > 0$ , then  $F'(t) > 0$ , and finally  $F(t) > 0$ . This proves the lower inequality in (82), and therefore (80).  $\square$

**Corollary 11.6** (A one-line branch-point approximation). *With  $u = -1 - \log(-x) > 0$ , define*

$$A_{-1}(x) = -1 - \sqrt{2u} - \frac{5}{6}u. \quad (83)$$

*Then*

$$|W_{-1}(x) - A_{-1}(x)| < \frac{u}{6}. \quad (84)$$

*Proof.* The approximation is the midpoint of the two bounds in (80); their distance is  $u/3$ .  $\square$

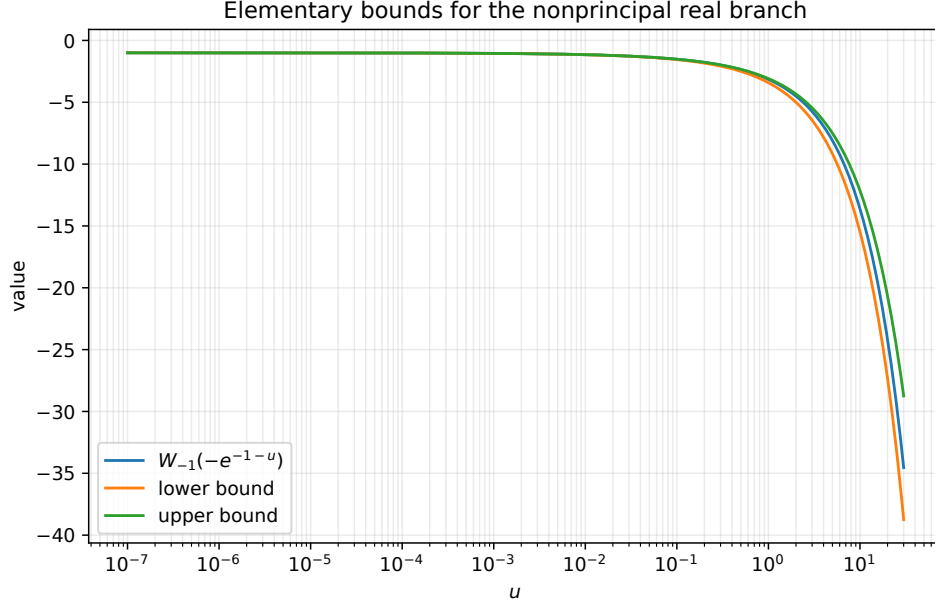


Figure 6: The two elementary bounds of [Theorem 11.5](#). They are especially effective near  $u = 0$ , which is the branch point.

A simpler square-root consequence, useful when only  $x$  is available, is

$$W_{-1}(x) < -1 - \sqrt{2(1 + ex)}.$$

Indeed, with  $w = -1 - v$ ,

$$v^2 - 2[1 - (1 + v)e^{-v}]$$

has derivative  $2v(1 - e^{-v}) > 0$  and vanishes at  $v = 0$ .

#### 11.4 Bounds for $W_{-1}$ near zero

The preceding  $u$ -bounds capture the branch point. A different pair captures the logarithmic singularity.

**Theorem 11.7** (Near-zero logarithmic bounds). *Let  $-1/e < x < 0$ , and put*

$$\eta = \log \frac{1}{-x} > 1.$$

*Then*

$$-\eta - \frac{\eta \log \eta}{\eta - 1} \leq W_{-1}(x) < -\eta - \log \eta. \quad (85)$$

*Proof.* Let  $y = -W_{-1}(x) > 1$ . Then

$$y - \log y = \eta, \quad \text{or} \quad y = \eta + \log y.$$

Since  $y > \eta$ , one has

$$y > \eta + \log \eta,$$

which gives the upper bound for  $W_{-1} = -y$ .

For the other direction, set

$$B = \eta + \frac{\eta \log \eta}{\eta - 1} = \eta + r, \quad r = \frac{\eta \log \eta}{\eta - 1}.$$

Because  $\log(1 + s) \leq s$ ,

$$\log B = \log \eta + \log \left(1 + \frac{r}{\eta}\right) \leq \log \eta + \frac{r}{\eta} = r.$$

Thus  $B - \log B \geq \eta$ . The function  $z - \log z$  is strictly increasing for  $z > 1$ , and  $y - \log y = \eta$ , so  $y \leq B$ . Multiplying by  $-1$  proves the lower bound in (85).  $\square$

The interval width in (85) is

$$\frac{\log \eta}{\eta - 1},$$

which tends to zero. Thus the two leading terms  $-\eta - \log \eta$  are not merely asymptotically plausible; they are enclosed by an elementary shrinking bracket.

## 12 Rational and continued-fraction approximations

### 12.1 Pade approximants at the origin

A  $[r/s]$  Pade approximant is a rational function  $P_r(x)/Q_s(x)$  whose Taylor series agrees with the target through as many terms as possible, with  $Q_s(0) = 1$ . Solving the resulting linear coefficient equations for  $W_0$  gives

$$R_{1,1}(x) = \frac{x}{1+x}, \tag{86}$$

$$R_{2,2}(x) = \frac{2x(4x+3)}{5x^2+14x+6}, \tag{87}$$

$$R_{3,3}(x) = \frac{3x(451x^2+912x+340)}{665x^3+3579x^2+3756x+1020}. \tag{88}$$

**Proposition 12.1** (Orders of contact). *At the origin,*

$$W_0(x) - R_{1,1}(x) = \frac{1}{2}x^3 + \mathcal{O}(x^4), \tag{89}$$

$$W_0(x) - R_{2,2}(x) = \frac{17}{72}x^5 + \mathcal{O}(x^6), \tag{90}$$

$$W_0(x) - R_{3,3}(x) = \frac{13489}{122400}x^7 + \mathcal{O}(x^8). \tag{91}$$

*None of these three rational functions has a pole on the real domain  $[-1/e, \infty)$  of  $W_0$ .*

*Proof.* Expand each denominator as a geometric power series at zero, multiply by its numerator, and compare with (40). This gives the displayed first unmatched terms. For  $R_{1,1}$ , the only pole is  $-1$ . The roots of the quadratic denominator in  $R_{2,2}$  are

$$\frac{-7 \pm \sqrt{19}}{5},$$

and both are less than  $-1/e$ .

For the cubic denominator, write

$$D(x) = 665x^3 + 3579x^2 + 3756x + 1020.$$

The exponential series gives  $e > 8/3$ , hence  $-1/e > -3/8$ . On  $[-3/8, \infty)$ , the quadratic

$$\frac{D'(x)}{3} = 665x^2 + 2386x + 1252$$

is increasing, because its derivative  $1330x + 2386$  is positive there; its value at  $-3/8$  is  $28849/64 > 0$ . Thus  $D$  is increasing on that interval, and

$$D(-3/8) = \frac{40821}{512} > 0.$$

It follows that  $D(x) > 0$  for every  $x \geq -1/e$ . Hence the cubic approximant has no pole on the principal real domain.  $\square$

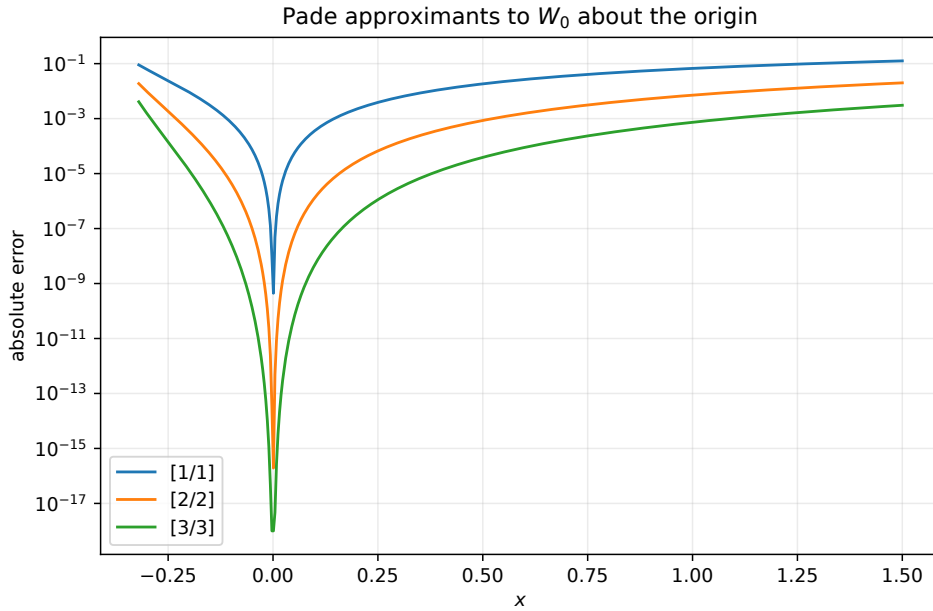


Figure 7: Absolute errors of three diagonal Pade approximants. Their main advantage over a Taylor polynomial is sensible behavior beyond the disk  $|x| < 1/e$ .

These approximants belong naturally to  $W_0$ . The branch  $W_{-1}$  has no Taylor series at  $x = 0$ ; for it, one should first use the branch-point variable  $p$ , the exponential variable  $t$ , or the logarithmic variable  $\eta$ , and then approximate the transformed function.

## 12.2 Euler's continued fraction

For a generalized continued fraction, write

$$b_0 + \mathbf{K}_{n=1}^N \frac{\alpha_n}{\beta_n} = b_0 + \frac{\alpha_1}{\beta_1 + \frac{\alpha_2}{\beta_2 + \cdots + \frac{\alpha_N}{\beta_N}}}.$$

**Lemma 12.2** (Euler connection). *Let  $c_0, c_1, \dots$  be nonzero where ratios are used, and put  $r_n = c_n/c_{n-1}$ . The  $N$ -th convergent of*

$$c_0 + \frac{c_1}{1 + \mathbf{K}_{n=2}^N \frac{-r_n}{1+r_n}} \quad (92)$$

*is exactly the partial sum  $\sum_{n=0}^N c_n$ .*

*Proof.* The numerator and denominator continuants  $P_n, Q_n$  of a generalized continued fraction obey

$$P_n = \beta_n P_{n-1} + \alpha_n P_{n-2}, \quad Q_n = \beta_n Q_{n-1} + \alpha_n Q_{n-2}.$$

This recurrence follows by clearing the last denominator in the nested fraction. Here

$$\alpha_1 = c_1, \quad \beta_1 = 1, \quad \alpha_n = -r_n, \quad \beta_n = 1 + r_n \quad (n \geq 2),$$

with  $P_0 = c_0, Q_0 = 1$ . The first convergent is  $P_1/Q_1 = c_0 + c_1$ . If  $P_{n-1} = \sum_{j=0}^{n-1} c_j$ ,  $P_{n-2} = \sum_{j=0}^{n-2} c_j$ , and  $Q_{n-1} = Q_{n-2} = 1$ , then

$$P_n = (1 + r_n) \sum_{j=0}^{n-1} c_j - r_n \sum_{j=0}^{n-2} c_j = \sum_{j=0}^{n-1} c_j + r_n c_{n-1} = \sum_{j=0}^n c_j,$$

while  $Q_n = (1 + r_n) - r_n = 1$ . Induction proves the claim.  $\square$

For the Lambert series, let

$$c_n = \frac{(-n)^{n-1}}{n!} x^n \quad (n \geq 1).$$

A calculation gives

$$\frac{c_n}{c_{n-1}} = - \left(1 - \frac{1}{n}\right)^{2-n} x =: a_n x.$$

**Theorem 12.3** (Euler continued fraction for  $W_0$ ). *For  $|x| < 1/e$ ,*

$$\boxed{W_0(x) = \frac{x}{1 + \mathbf{K}_{n=2}^\infty \frac{-a_n x}{1+a_n x}}, \quad a_n = - \left(1 - \frac{1}{n}\right)^{2-n}.} \quad (93)$$

*The convergents are exactly the partial sums of the Maclaurin series.*

*Proof.* Apply [Theorem 12.2](#) with  $c_0 = 0$  and the displayed ratios. Each convergent equals a Maclaurin partial sum, and those partial sums converge to  $W_0(x)$  for  $|x| < 1/e$ .  $\square$

*Literature note 12.4.* The paper of Corcino, Corcino, and Mezó in the supplied archive develops more sophisticated regular and associated continued fractions, including coefficient formulas through Hankel determinants and numerical studies of their convergence. Formula (93) is the elementary Euler connection from which that discussion begins.

## 13 Contour and real integral representations

Integral representations are useful for proofs of monotonicity, for quadrature, and for connecting  $W$  with complex analysis. We begin with a formula that applies to every branch.

### 13.1 A local contour representation for all branches

**Theorem 13.1** (Inverse-function residue formula). *Let  $z$  be a regular value of  $f(\zeta) = \zeta e^\zeta$ , and let  $\Gamma_k$  be a small positively oriented contour enclosing the one simple root  $\zeta = W_k(z)$  and no other root of  $f(\zeta) = z$ . Then, for every integer  $m \geq 0$ ,*

$$W_k(z)^m = \frac{1}{2\pi i} \int_{\Gamma_k} \frac{\zeta^m e^\zeta (1 + \zeta)}{\zeta e^\zeta - z} d\zeta. \quad (94)$$

For  $m = 1$ , this is an integral representation of  $W_k(z)$  itself.

*Proof.* The denominator has a simple zero at  $\zeta = W_k(z)$ , because  $f'(\zeta) = e^\zeta(1 + \zeta) \neq 0$  at a regular value. The residue of the integrand there is

$$\left. \frac{\zeta^m f'(\zeta)}{f'(\zeta)} \right|_{\zeta=W_k(z)} = W_k(z)^m.$$

The residue theorem proves the formula. □

This representation includes  $W_{-1}$  whenever a contour isolates that preimage. What follows is more special: a positive real integral for the principal branch, in the family studied by Kalugin, Jeffrey, and Corless [11].

### 13.2 The principal-branch cut parametrization

For  $0 < v < \pi$ , define

$$a(v) = v e^{-v \cot v} \csc v. \quad (95)$$

**Lemma 13.2** (Properties of  $a(v)$ ). *The function  $a$  is strictly increasing and maps  $(0, \pi)$  onto  $(1/e, \infty)$ . Moreover,*

$$\frac{v a'(v)}{a(v)} = 1 - 2v \cot v + v^2 \csc^2 v = (1 - v \cot v)^2 + v^2 > 0. \quad (96)$$

*Proof.* Taking logarithms of (95) and differentiating gives

$$\frac{a'}{a} = \frac{1}{v} - 2 \cot v + v \csc^2 v,$$

which is (96). Its final expression is positive.

As  $v \rightarrow 0$ , the expansions  $\sin v \sim v$  and  $v \cot v \rightarrow 1$  give  $a(v) \rightarrow 1/e$ . As  $v \rightarrow \pi^-$ , one has  $\csc v \rightarrow +\infty$  and  $-v \cot v \rightarrow +\infty$ , so  $a(v) \rightarrow \infty$ . Strict monotonicity now gives the claimed range. □

Why does this function appear? On the upper edge of the principal branch cut, write

$$W_0(-a + i0) = u + iv, \quad 0 < v < \pi.$$

The condition that  $(u + iv)e^{u+iv}$  be negative real forces

$$u = -v \cot v,$$

and its real value is

$$-v e^{-v \cot v} \csc v = -a(v).$$

Thus  $v$  parametrizes the entire cut from  $-1/e$  to  $-\infty$ .



**Theorem 13.3** (Stieltjes and logarithmic integrals). *For  $z \in \mathbb{C} \setminus (-\infty, -1/e]$ ,*

$$W_0'(z) = \frac{1}{\pi} \int_0^\pi \frac{dv}{z + a(v)}, \quad (97)$$

$$W_0(z) = \frac{1}{\pi} \int_0^\pi \operatorname{Log} \left( 1 + \frac{z}{a(v)} \right) dv, \quad (98)$$

$$W_0(z) = \frac{z}{\pi} \int_0^\pi \frac{(1 - v \cot v)^2 + v^2}{z + a(v)} dv. \quad (99)$$

*For real  $z > -1/e$ , all three integrals are real and the ordinary real logarithm may be used in (98).*

*Proof.* We first prove (97). Let  $F(z) = W_0'(z)$ . It is analytic off the cut, behaves as  $\mathcal{O}(1/z)$  at infinity, and has an integrable inverse-square-root singularity at  $-1/e$ . On the upper edge of the cut,

$$W_0(-a(v) + i0) = -v \cot v + iv.$$

Differentiating with respect to  $v$ , while  $z = -a(v)$ , gives

$$F(-a(v) + i0) = -\frac{(-v \cot v)' + i}{a'(v)}.$$

The lower boundary value is its complex conjugate. Hence the jump is

$$F_+ - F_- = -\frac{2i}{a'(v)}. \quad (100)$$

Apply Cauchy's formula on a large keyhole contour surrounding the cut. The outer circle contributes zero in the limit because  $F(z) = \mathcal{O}(1/z)$ ; a small circle around the branch point contributes zero because its length is order  $r$  and  $F$  is order  $r^{-1/2}$ . The two sides of the cut therefore give

$$F(z) = \frac{1}{2\pi i} \int_{-\infty}^{-1/e} \frac{F_+(t) - F_-(t)}{t - z} dt.$$

Set  $t = -a(v)$ . As  $t$  runs from  $-\infty$  to  $-1/e$ ,  $v$  runs from  $\pi$  to 0. Using (100), the integral becomes

$$F(z) = \frac{1}{\pi} \int_0^\pi \frac{dv}{z + a(v)},$$

as claimed.

Integrate this identity from 0 to  $z$  along any path in the slit plane. Since  $W_0(0) = 0$ , and since differentiation under the integral is justified on compact subsets away from the cut,

$$W_0(z) = \frac{1}{\pi} \int_0^\pi \left( \int_0^z \frac{ds}{s + a(v)} \right) dv = \frac{1}{\pi} \int_0^\pi \operatorname{Log} \left( 1 + \frac{z}{a(v)} \right) dv.$$

This proves (98).

Finally let  $h(v) = \operatorname{Log}(1 + z/a(v))$ . The boundary term  $[vh(v)]_0^\pi$  vanishes: the factor  $v$  handles  $v = 0$ , and  $a(v) \rightarrow \infty$  handles  $v = \pi$ . Integration by parts gives

$$\int_0^\pi h(v) dv = z \int_0^\pi \frac{va'(v)}{a(v)[z + a(v)]} dv.$$

Insert (96) to obtain (99). □

**Corollary 13.4** (Alternating signs of all principal-branch derivatives). *For every integer  $n \geq 0$  and every real  $x > -1/e$ ,*

$$(-1)^n W_0^{(n+1)}(x) = \frac{n!}{\pi} \int_0^\pi \frac{dv}{[x + a(v)]^{n+1}} > 0. \quad (101)$$

*Thus  $W_0'$  is completely monotone:  $W_0' > 0$ ,  $W_0'' < 0$ ,  $W_0''' > 0$ , and so on.*

*Proof.* Differentiate (97)  $n$  times under the integral. For  $x > -1/e$ , Theorem 13.2 gives  $x + a(v) > 0$ , so the integrand on the right of (101) is positive.  $\square$

This all-orders sign property is special to the principal branch. It cannot extend to  $W_{-1}$ , whose second derivative changes sign at  $-2/e^2$  by Theorem 5.3.

**Corollary 13.5** (An integral for rooted-tree numbers). *For every integer  $m \geq 1$ ,*

$$\frac{1}{\pi} \int_0^\pi a(v)^{-m} dv = \frac{m^{m-1}}{(m-1)!}. \quad (102)$$

*Proof.* Set  $x = 0$  and  $n = m - 1$  in (101). The left side is

$$(-1)^{m-1} W_0^{(m)}(0) = m^{m-1}$$

by (42). Divide by  $(m-1)!$ .  $\square$

Expanding the logarithm in (98) and using (102) gives a second proof of the Maclaurin series:

$$\frac{1}{\pi} \int_0^\pi \sum_{m \geq 1} \frac{(-1)^{m-1}}{m} \frac{z^m}{a(v)^m} dv = \sum_{m \geq 1} \frac{(-m)^{m-1}}{m!} z^m.$$

This identity ties together three representations that at first seem unrelated: rooted labelled trees, a branch-cut parametrization, and the inverse of  $we^w$ .

## 14 Numerical evaluation

A computer does not evaluate a special function by magic. It combines:

1. a representation adapted to the part of the domain,
2. a starting approximation,
3. a rapidly convergent correction,
4. and a reliable stopping or error test.

The branch point, the origin, and large arguments require different representations.

### 14.1 Residuals give certified a posteriori errors

**Theorem 14.1** (Residual error bound). *Let  $r = W_k(x)$ , let  $\hat{w}$  be an approximation on the same real inverse interval, and let  $I$  be the closed interval joining  $r$  and  $\hat{w}$ . If  $I$  does not contain  $-1$ , put*

$$m_I = \min_{t \in I} |e^t(1+t)|.$$

*Then*

$$|\hat{w} - r| \leq \frac{|\hat{w}e^{\hat{w}} - x|}{m_I}. \quad (103)$$

For  $x \neq 0$ , an overflow-resistant alternative uses

$$G_x(w) = w + \log |w| - \log |x|.$$

If, in addition,  $I$  avoids zero and

$$\mu_I = \min_{t \in I} |1 + 1/t| > 0,$$

then

$$|\hat{w} - r| \leq \frac{|G_x(\hat{w})|}{\mu_I}. \quad (104)$$

*Proof.* Apply the mean value theorem to  $f(t) = te^t - x$ :

$$f(\hat{w}) - f(r) = f'(\xi)(\hat{w} - r)$$

for some  $\xi \in I$ . Since  $f(r) = 0$  and  $|f'(\xi)| \geq m_I$ , (103) follows. The proof of (104) is identical, using  $G_x(r) = 0$  and  $G'_x(t) = 1 + 1/t$ .  $\square$

At the branch point  $m_I$  and  $\mu_I$  can be small. This is not a defect of the estimate: Equation (24) shows that the inverse problem itself is ill-conditioned there. The signed variable  $p$  is the correct coordinate in that regime.

## 14.2 Newton and Halley iterations

For

$$f_x(w) = we^w - x,$$

Newton's method is

$$w^+ = w - \frac{we^w - x}{e^w(1 + w)} = \frac{w^2 + xe^{-w}}{1 + w}. \quad (105)$$

Halley's method is

$$w^+ = w - \frac{2f_x(w)f'_x(w)}{2f'_x(w)^2 - f_x(w)f''_x(w)}, \quad f''_x(w) = e^w(w + 2). \quad (106)$$

**Theorem 14.2** (Local orders of convergence). *Let  $r$  be a simple root of a three-times continuously differentiable function  $f$ , so  $f(r) = 0$  and  $f'(r) \neq 0$ . If  $w = r + e$  is sufficiently close to  $r$ , Newton's method has*

$$e^+ = \frac{f''(r)}{2f'(r)}e^2 + \mathcal{O}(e^3),$$

*while Halley's method has  $e^+ = \mathcal{O}(e^3)$ . Applied to the Lambert equation away from  $r = -1$ , Newton is locally quadratic and Halley locally cubic.*

*Proof.* Write  $a = f'(r)$ ,  $b = f''(r)$ ,  $c = f'''(r)$ . Taylor expansion gives

$$f(r + e) = ae + \frac{b}{2}e^2 + \frac{c}{6}e^3 + \mathcal{O}(e^4),$$

$$f'(r + e) = a + be + \frac{c}{2}e^2 + \mathcal{O}(e^3).$$

Division yields

$$\frac{f(r + e)}{f'(r + e)} = e - \frac{b}{2a}e^2 + \mathcal{O}(e^3),$$

which proves the Newton formula.

For Halley, substitute the same expansions into

$$\frac{2ff'}{2(f')^2 - ff''}.$$

Both numerator and denominator have the same relative linear term, so their quotient is  $e + \mathcal{O}(e^3)$ ; the  $e^2$  term cancels. Subtracting this correction from  $e$  leaves  $\mathcal{O}(e^3)$ .  $\square$

The root at the common branch point is not simple:  $f'(-1) = 0$ . One should return  $-1$  exactly when  $x = -1/e$ , or use the square-root expansion for nearby inputs.

### 14.3 Logarithmic Newton iteration

For nonzero  $x$ , solve instead

$$G_x(w) = w + \log |w| - \log |x| = 0.$$

Newton's formula simplifies to

$$N_x(w) = \frac{w}{1+w} \left( 1 + \log \left| \frac{x}{w} \right| \right). \quad (107)$$

This form is central in guaranteed high-precision algorithms because it avoids forming  $e^w$ .

We first prove a global one-step ordering theorem.

**Theorem 14.3** (Branch-safe monotone Newton step). *Let  $r = W_k(x)$ , and suppose  $w$  has the same sign as  $x$ .*

- (i) *If  $0 < w < r$ , then  $w < N_x(w) < r$ .*
- (ii) *If  $-1 < r < w < 0$ , then  $r < N_x(w) < w$ .*
- (iii) *If  $w < r < -1$ , then  $w < N_x(w) < r$ .*

*Proof.* Put  $t = r/w > 0$ . Since  $\log |x| = r + \log |r|$ , direct algebra gives the exact identities

$$N_x(w) - w = \frac{w}{1+w} [w(t-1) + \log t], \quad (108)$$

$$r - N_x(w) = \frac{w}{1+w} [t-1 - \log t]. \quad (109)$$

For every  $t > 0$ ,  $t \neq 1$ ,

$$\frac{t-1}{t} < \log t < t-1, \quad (110)$$

which is [Theorem 11.1](#) after a change of variable.

In case (i),  $t > 1$ ,  $w/(1+w) > 0$ , and both brackets in (108) and (109) are positive. Hence  $w < N_x(w) < r$ .

In case (ii),  $t > 1$  and  $w/(1+w) < 0$ . Since  $r = wt > -1$ ,  $w > -1/t$ , and therefore

$$w(t-1) + \log t > -\frac{t-1}{t} + \log t > 0.$$

Thus (108) gives  $N_x(w) < w$ , while (109) gives  $r < N_x(w)$ .

In case (iii),  $0 < t < 1$  and  $w/(1+w) > 0$ . Since  $r = wt < -1$ , one has  $w < -1/t$ . Multiplying by the negative number  $t-1$  gives

$$w(t-1) > \frac{1-t}{t}.$$

The left inequality of (110) is  $\log t > -(1-t)/t$ , so the bracket in (108) is positive. The bracket in (109) is always positive. Hence  $w < N_x(w) < r$ .  $\square$

**Theorem 14.4** (Elementary globally convergent starts). *For a real input in the stated branch domain, choose*

$$w_0 = \begin{cases} x/(1+x), & k=0, \quad 0 < x < e, \\ \log x - \log \log x, & k=0, \quad x \geq e, \\ x, & k=0, \quad -1/e < x < 0, \\ -1 - \sqrt{2u} - u, & k=-1, \quad -1/e < x < 0, \quad u = -1 - \log(-x). \end{cases} \quad (111)$$

Then  $w_{n+1} = N_x(w_n)$  is well defined, remains on the selected branch interval, is monotone, and converges to  $W_k(x)$ .

*Proof.* For  $0 < x < e$ , [Theorem 11.2](#) gives  $0 < w_0 < W_0(x)$ , so case (i) of [Theorem 14.3](#) applies. For  $x > e$ , [Theorem 11.3](#) gives the same ordering; at  $x = e$ , the start is the exact value 1.

For  $-1/e < x < 0$  on  $W_0$ , (79) gives  $-1 < W_0(x) < w_0 = x < 0$ , so case (ii) applies. On  $W_{-1}$ , [Theorem 11.5](#) gives  $w_0 < W_{-1}(x) < -1$ , so case (iii) applies.

In each case the sequence is monotone and bounded by the root, so it has a limit  $\ell$  in the same interval. Continuity gives  $N_x(\ell) = \ell$ . Since  $\ell \neq 0, -1$ , this fixed-point equation reduces to

$$\log |x/\ell| = \ell,$$

or  $x = \ell e^\ell$ . Uniqueness on the selected inverse interval gives  $\ell = W_k(x)$ .  $\square$

**Corollary 14.5** (Quadratic error constant). *Let  $r = W_k(x) \neq -1, 0$ , and let  $e_n = w_n - r$  for the logarithmic Newton iteration. Then*

$$e_{n+1} = -\frac{1}{2r(1+r)}e_n^2 + \mathcal{O}(e_n^3). \quad (112)$$

*Proof.* Apply the Newton error formula of [Theorem 14.2](#) to  $G'_x(w) = 1 + 1/w$  and  $G''_x(w) = -1/w^2$ .  $\square$

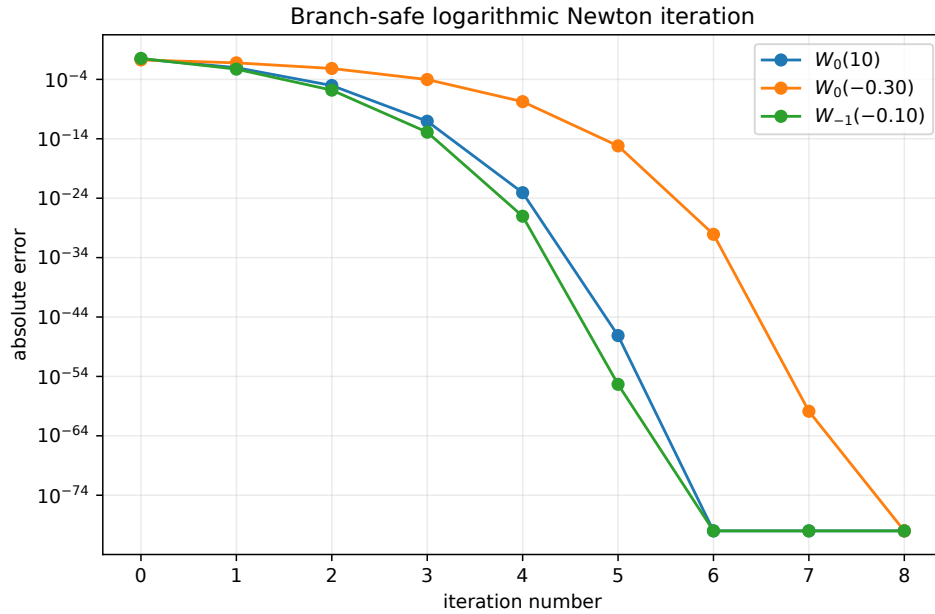


Figure 8: Monotone, eventually quadratic convergence of (107) in three different branch regimes.

## 14.4 A practical branch-aware algorithm

For ordinary floating-point work, the following strategy is robust.

**Algorithm 14.6** (Real Lambert  $W$ ). *Given a real  $x$  and branch  $k \in \{0, -1\}$ :*

1. *Reject inputs outside the branch domain. Return  $-1$  exactly at  $x = -1/e$ , and return 0 for  $W_0(0)$ .*
2. *If  $1 + ex$  is tiny, evaluate the signed Puiseux series (50).*
3. *If  $k = 0$  and  $|x|$  is small, use the Maclaurin series or a Pade approximant.*
4. *If  $k = 0$  and  $x$  is large, use (73) as a start. If  $k = -1$  and  $x$  is close to zero, use (74).*
5. *Otherwise use a guaranteed start from (111).*
6. *Apply logarithmic Newton or Halley until the logarithmic residual and the change in  $w$  are both below the requested tolerance.*

If the input  $x < 0$  is so small that it underflows, store  $\lambda = \log(-x)$  instead. Formula (107) then becomes

$$N_\lambda(w) = \frac{w}{1+w} (1 + \lambda - \log(-w)), \quad (113)$$

which never needs  $x$  itself.

*Literature note 14.7.* The 2022 paper by Lajos Lőczi [13] in the supplied archive develops guaranteed-precision versions of logarithmic Newton iteration, including piecewise starting values and explicit error estimates over the full real domains. The paper by Iacono and Boyd [10] constructs global analytic approximations and iteration-based refinements for  $W_0$ . Howard’s Schröder-series work [9] studies higher-order root-solving expansions. The present article emphasizes a shorter set of starts for which monotone convergence can be proved directly by Theorem 14.3.

## 14.5 Which approximation should be used?

| Regime                          | Recommended representation           | Natural small parameter |
|---------------------------------|--------------------------------------|-------------------------|
| $W_0, x \approx 0$              | Maclaurin series or Pade approximant | $x$                     |
| Either branch, $x \approx -1/e$ | signed Puiseux series                | $p = \pm\sqrt{2(1+ex)}$ |
| $W_0, x \gg 1$                  | logarithmic asymptotic series        | $(\log \log x)/\log x$  |
| $W_{-1}, x \rightarrow 0^-$     | logarithmic asymptotic series        | $\log \eta/\eta$        |
| Moderate $x$ , high accuracy    | logarithmic Newton or Halley         | current residual        |
| Certified value                 | monotone bracket plus residual bound | bracket width           |

There is no single best formula. A high-quality implementation is a carefully organized collection of mathematically equivalent formulas, each used where it is well conditioned.

## 15 Solving transcendental equations

The main skill is to isolate an expression of the form

$$Ye^Y = Z.$$

Branch selection is then decided by the allowable range of the original unknown, not by algebra alone.

## 15.1 A linear term plus a logarithm

**Theorem 15.1.** *Let  $a, b, c \in \mathbb{R}$ , with  $ab \neq 0$ , and seek  $x > 0$  satisfying*

$$ax + b \log x = c.$$

*Every real solution is*

$$x = \frac{b}{a} W_k \left( \frac{a}{b} e^{c/b} \right), \quad (114)$$

*where a real branch is chosen so that the right side is positive.*

*Proof.* Rearrange and exponentiate:

$$x = e^{c/b} e^{-(a/b)x}.$$

Therefore

$$\frac{a}{b} x \exp \left( \frac{a}{b} x \right) = \frac{a}{b} e^{c/b}.$$

Apply a real branch of  $W$ , then multiply by  $b/a$ . Every reversible step preserves all positive solutions.  $\square$

Two important special cases are

$$x + \log x = c \implies x = W_0(e^c),$$

and

$$x - \log x = c \implies x = -W_k(-e^{-c}). \quad (115)$$

For  $c > 1$ , (115) has two positive solutions:  $-W_0(-e^{-c}) \in (0, 1)$  and  $-W_{-1}(-e^{-c}) \in (1, \infty)$ . At  $c = 1$ , they coalesce at  $x = 1$ . This is exactly the two-branch geometry of [Equation \(5\)](#).

## 15.2 A power times an exponential

**Theorem 15.2.** *Let  $p \neq 0$ ,  $q \neq 0$ , and suppose a consistent real  $p$ -th root of  $A$  has been selected. Solutions of*

$$x^p e^{qx} = A$$

*that are compatible with that root are*

$$x = \frac{p}{q} W_k \left( \frac{q}{p} A^{1/p} \right). \quad (116)$$

*Proof.* Taking the selected  $p$ -th root gives  $x e^{(q/p)x} = A^{1/p}$ . Multiplication by  $q/p$  produces the defining Lambert equation. Conversely, substitution verifies every value in (116).  $\square$

When  $p$  is even, taking a real root can introduce two signs. Each sign must be treated separately, and each resulting Lambert argument may itself admit two real branches.

### 15.3 An exponential equal to an affine function

**Theorem 15.3.** *Let  $\alpha\beta \neq 0$ . Every solution of*

$$e^{\alpha x} = \beta x + \gamma \quad (117)$$

*is*

$$\boxed{x = -\frac{\gamma}{\beta} - \frac{1}{\alpha} W_k \left( -\frac{\alpha}{\beta} e^{-\alpha\gamma/\beta} \right)}. \quad (118)$$

*Real solutions correspond exactly to real branches available at the displayed argument.*

*Proof.* Set  $y = \beta x + \gamma$ , so  $x = (y - \gamma)/\beta$ . Equation (117) becomes

$$e^{\alpha(y-\gamma)/\beta} = y.$$

Multiplying by  $\exp(-\alpha y/\beta)$  and then by  $-\alpha/\beta$  gives

$$\left( -\frac{\alpha y}{\beta} \right) \exp \left( -\frac{\alpha y}{\beta} \right) = -\frac{\alpha}{\beta} e^{-\alpha\gamma/\beta}.$$

Apply  $W_k$ , solve for  $y$ , and then for  $x$ . □

**Example 15.4.** *The equation  $e^x = 3x$  has argument  $-1/3 \in (-1/e, 0)$ , so it has two positive solutions:*

$$x = -W_0(-1/3), \quad x = -W_{-1}(-1/3).$$

*The two intersections of an exponential and a line are precisely the two real Lambert branches.*

### 15.4 The equation $x^x = A$

**Theorem 15.5** (All positive solutions of  $x^x = A$ ). *Let  $A > 0$ . Positive solutions of  $x^x = A$  are*

$$x = \exp(W_k(\log A)). \quad (119)$$

*Their number is:*

- *one if  $A > 1$ ;*
- *one if  $A = 1$ , namely  $x = 1$ ;*
- *two if  $e^{-1/e} < A < 1$ ;*
- *one double minimum solution if  $A = e^{-1/e}$ , namely  $x = e^{-1}$ ;*
- *none if  $0 < A < e^{-1/e}$ .*

*Proof.* Put  $y = \log x$ , so  $x = e^y$ . Taking logarithms of  $x^x = A$  gives

$$ye^y = \log A.$$

Thus  $y = W_k(\log A)$ , proving (119). If  $\log A > 0$ , only  $W_0$  is real. If  $-1/e < \log A < 0$ , both  $W_0$  and  $W_{-1}$  are real. At  $\log A = -1/e$ , they meet. Below that value no real branch exists.

For a direct calculus check, the derivative of  $\log(x^x) = x \log x$  is  $1 + \log x$ , so  $x^x$  has its positive minimum at  $x = e^{-1}$ , with value  $e^{-1/e}$ . □



## 15.5 Fixed points of exponentiation

A positive fixed point of

$$y = x^y$$

satisfies

$$y = -\frac{W_k(-\log x)}{\log x} \quad (x \neq 1). \quad (120)$$

*Proof.* Taking logarithms gives  $\log y = y \log x$ . Set  $z = -y \log x$ . Then

$$ze^z = (-y \log x)e^{-y \log x} = (-y \log x)\frac{1}{y} = -\log x.$$

Thus  $z = W_k(-\log x)$ , which gives (120).  $\square$

For  $0 < x < 1$ , the argument  $-\log x$  is positive and there is one positive fixed point. For  $1 < x < e^{1/e}$ , the argument lies in  $(-1/e, 0)$ , so there are two positive fixed points. At  $x = e^{1/e}$ , they merge at  $y = e$ . This statement concerns solutions of the fixed-point equation; convergence of the iterated power tower is a separate dynamical question.

## 15.6 A Planck-type maximization problem

**Theorem 15.6.** *Let  $n > 1$ . The function*

$$F_n(x) = \frac{x^n}{e^x - 1}, \quad x > 0,$$

*has a unique critical point, which is its global maximum, at*

$$x_n = n + W_0(-ne^{-n}). \quad (121)$$

*The second real branch gives*

$$n + W_{-1}(-ne^{-n}) = 0,$$

*the boundary point rather than an interior maximum.*

*Proof.* Logarithmic differentiation gives

$$\frac{F'_n(x)}{F_n(x)} = \frac{n}{x} - \frac{e^x}{e^x - 1}.$$

A critical point satisfies

$$n(e^x - 1) = xe^x,$$

or

$$(n - x)e^x = n.$$

Put  $y = x - n$ . Then

$$ye^y = -ne^{-n},$$

so  $x = n + W_k(-ne^{-n})$ . Since  $-n \leq -1$ , [Theorem 2.3](#) gives  $W_{-1}(-ne^{-n}) = -n$ , hence the boundary value  $x = 0$ . The principal value lies in  $(-1, 0)$ , so  $x_n \in (n - 1, n)$ .

Finally  $F_n(x) \rightarrow 0$  both as  $x \downarrow 0$  (because  $n > 1$ ) and as  $x \rightarrow \infty$ . The critical-point equation has only the one interior solution just found, so that solution is the global maximum.  $\square$

## 16 Applications

The point of a special function is not merely to rename an equation. A closed form exposes parameter dependence, branch multiplicity, asymptotic regimes, and reliable initial values for computation.

### 16.1 Michaelis–Menten substrate depletion

Consider

$$\frac{dS}{dt} = -\frac{V_{\max}S}{K_M + S}, \quad S(0) = S_0 > 0. \quad (122)$$

**Theorem 16.1.** *The solution is*

$$S(t) = K_M W_0 \left( \frac{S_0}{K_M} \exp \left[ \frac{S_0}{K_M} - \frac{V_{\max}t}{K_M} \right] \right). \quad (123)$$

*Proof.* Separate variables:

$$\left( 1 + \frac{K_M}{S} \right) dS = -V_{\max} dt.$$

Integration and the initial condition give

$$S + K_M \log S = S_0 + K_M \log S_0 - V_{\max}t.$$

Put  $y = S/K_M$  and  $y_0 = S_0/K_M$ . Constants involving  $\log K_M$  cancel, leaving

$$y + \log y = y_0 + \log y_0 - \frac{V_{\max}t}{K_M}.$$

Exponentiation gives

$$ye^y = y_0 \exp \left( y_0 - \frac{V_{\max}t}{K_M} \right).$$

The right side is positive, so the only real branch is  $W_0$ , proving (123).  $\square$

The archive paper by Goličnik [8] develops this and related biochemical kinetic examples in detail. Formula (123) makes it possible to differentiate with respect to parameters without repeatedly solving an implicit equation.

### 16.2 A diode with a series resistor

Let  $I_S > 0$  be the saturation current,  $R_S > 0$  the series resistance, and  $nV_T > 0$  the thermal-voltage scale. The Shockley law and the circuit voltage relation are

$$I = I_S \left( e^{V_D/(nV_T)} - 1 \right), \quad V = V_D + IR_S.$$

**Theorem 16.2.** *The current is*

$$I = \frac{nV_T}{R_S} W_0 \left( \frac{R_S I_S}{nV_T} \exp \left[ \frac{V + R_S I_S}{nV_T} \right] \right) - I_S. \quad (124)$$

*Proof.* Let  $Y = I + I_S$ . Substitution of  $V_D = V - IR_S$  gives

$$Y = I_S \exp \left( \frac{V - R_S(Y - I_S)}{nV_T} \right).$$

Therefore

$$Y \exp\left(\frac{R_S Y}{nV_T}\right) = I_S \exp\left(\frac{V + R_S I_S}{nV_T}\right).$$

Multiplication by  $R_S/(nV_T)$  produces a positive Lambert argument, so  $W_0$  is the unique real branch. Solving for  $I = Y - I_S$  gives (124).  $\square$

This branch-explicit form underlies the transistor and diode analyses in Banwell's paper [1].

### 16.3 Characteristic exponents of a delay equation

Consider

$$y'(t) = a y(t - \tau), \quad \tau > 0. \quad (125)$$

**Theorem 16.3.** *Exponential modes  $y(t) = e^{\lambda t}$  have exponents*

$$\lambda_k = \frac{1}{\tau} W_k(a\tau), \quad k \in \mathbb{Z}. \quad (126)$$

*Proof.* Substitution gives

$$\lambda e^{\lambda t} = a e^{\lambda(t-\tau)},$$

so  $\lambda = a e^{-\lambda\tau}$ . Hence

$$(\lambda\tau)e^{\lambda\tau} = a\tau,$$

and (126) follows.  $\square$

Here the infinitely many complex branches are not optional decoration: they form the characteristic spectrum. If  $-1/e < a\tau < 0$ , the two real branches give two real characteristic exponents in addition to the complex ones.

### 16.4 The final-size equation of an epidemic model

A common normalized SIR final-size equation is

$$\log \frac{s_\infty}{s_0} = -R_0(1 - s_\infty), \quad (127)$$

where  $R_0 > 0$  and  $0 < s_\infty \leq s_0 \leq 1$ .

**Theorem 16.4.** *Assume  $R_0 > 0$ . Every positive algebraic solution is*

$$s_\infty = -\frac{1}{R_0} W_k(-R_0 s_0 e^{-R_0}). \quad (128)$$

*The physically admissible branch must also satisfy  $s_\infty \leq s_0$  and the assumptions under which (127) was derived.*

*Proof.* Exponentiating (127) gives

$$s_\infty = s_0 e^{-R_0} e^{R_0 s_\infty}.$$

Thus

$$(-R_0 s_\infty) e^{-R_0 s_\infty} = -R_0 s_0 e^{-R_0},$$

which proves (128).  $\square$

When two real branches are available, one may be a mathematically valid but physically inadmissible root. For example, if  $s_0 = 1$  and  $R_0 > 1$ , the identity

$$W_{-1}(-R_0 e^{-R_0}) = -R_0$$

gives  $s_\infty = 1$ , the no-outbreak root, while  $W_0$  gives the smaller nontrivial root. Initial conditions and model interpretation decide which one is relevant.

## 16.5 Inverting the prime-number scale

The rough relation

$$n \approx \frac{x}{\log x}$$

arises when inverting the prime number theorem. For  $n \geq e$ , solving the equality exactly and selecting its large solution gives

$$x = -nW_{-1}(-1/n). \quad (129)$$

*Proof.* Put  $y = \log x$ , so  $x = e^y$ . The equation  $x/\log x = n$  becomes

$$\frac{e^y}{y} = n,$$

or

$$(-y)e^{-y} = -\frac{1}{n}.$$

The large solution has  $y > 1$ , so  $-y < -1$ , which selects  $W_{-1}$ :  $y = -W_{-1}(-1/n)$ . Exponentiating and using  $e^{-W(z)} = W(z)/z$  gives  $x = -nW_{-1}(-1/n)$ .  $\square$

The principal branch would give a small root near  $x = 1$ , not the large- $x$  inverse relevant to primes. Formula (74) immediately yields

$$-nW_{-1}(-1/n) = n \left( \log n + \log \log n + \frac{\log \log n}{\log n} + \dots \right),$$

illustrating why the nonprincipal branch naturally enters prime asymptotics. The archive paper by Visser develops this viewpoint for bounds on the  $n$ -th prime.

## 17 Series in a logarithmic argument

For positive arguments it is sometimes better to expand the Wright omega function

$$\omega(t) = W_0(e^t)$$

than to expand  $W_0(x)$  directly. Its defining equation

$$\omega + \log \omega = t$$

has no exponentially large quantity.

**Theorem 17.1** (Expansion centered at  $x = e$ ). *Let  $h = \log x - 1$ . For  $x > 0$  sufficiently close to  $e$ ,*

$$W_0(x) = 1 + \frac{h}{2} + \frac{h^2}{16} - \frac{h^3}{192} - \frac{h^4}{3072} + \frac{13h^5}{61440} + \mathcal{O}(h^6). \quad (130)$$

*Proof.* Write  $\omega = 1 + s$  and  $t = 1 + h$ . The defining equation becomes

$$h = s + \log(1 + s) = 2s - \frac{1}{2}s^2 + \frac{1}{3}s^3 - \frac{1}{4}s^4 + \cdots.$$

The derivative of the right side at  $s = 0$  is  $2 \neq 0$ , so it has an analytic local inverse. Substitute  $s = c_1 h + c_2 h^2 + \cdots$  and compare coefficients. Successive comparison gives

$$c_1 = \frac{1}{2}, \quad c_2 = \frac{1}{16}, \quad c_3 = -\frac{1}{192}, \quad c_4 = -\frac{1}{3072}, \quad c_5 = \frac{13}{61440},$$

which proves (130). □

This representation, discussed in the Schroeder-series material in the archive, is useful when an application supplies  $\log x$  directly. Its nonprincipal real analogue is not an ordinary Taylor series at the common endpoint: if

$$\nu(\eta) = -W_{-1}(-e^{-\eta}),$$

then  $\nu - \log \nu = \eta$  and the derivative with respect to  $\nu$  vanishes at  $(\eta, \nu) = (1, 1)$ . The correct local expansion is therefore the signed square-root series (58), not a Taylor series. This is another manifestation of the branch-point asymmetry.

## 18 A brief guide to the complex branches

A complete construction of the Riemann surface belongs to complex analysis, but the main ideas follow directly from the defining map.

### 18.1 Local branches and the critical value

The complex derivative of  $f(w) = we^w$  is

$$f'(w) = e^w(1 + w).$$

Thus  $w = -1$  is the only finite critical point, and its critical value is  $-1/e$ . At every other preimage  $w_*$ , the complex inverse function theorem gives a unique analytic local branch through  $w_*$ . Near  $-1/e$ , the same calculation as in Section 9 gives the complex square-root coordinate

$$p = \sqrt{2(1 + ez)}.$$

At  $z = 0$ , the principal branch is regular and satisfies  $W_0(0) = 0$ . The other branches escape to complex infinity as  $z \rightarrow 0$ , creating logarithmic branching.

### 18.2 Indexing and the all-branch asymptotic expansion

For nonzero  $z$ , choose a determination of  $\text{Log } z$  and define

$$L_{1,k} = \text{Log } z + 2\pi i k, \quad L_{2,k} = \text{Log } L_{1,k}.$$

For large  $|k|$ , a solution is close to  $L_{1,k} - L_{2,k}$ . The standard indexing  $W_k$  is chosen so that

$$W_k(z) = L_{1,k} - L_{2,k} + o(1) \quad (|k| \rightarrow \infty), \tag{131}$$

with consistent continuation away from branch cuts.

**Theorem 18.1** (Complex all-branch expansion). *Whenever  $|L_{1,k}| \rightarrow \infty$  while the logarithms are chosen consistently,*

$$W_k(z) \sim L_{1,k} - L_{2,k} + \sum_{n=1}^{\infty} \frac{\mathcal{P}_n(L_{2,k})}{L_{1,k}^n}, \quad (132)$$

*with the same Stirling polynomials  $\mathcal{P}_n$  as in (63).*

*Proof.* The complex equation is

$$W + \text{Log } W = L_{1,k}.$$

Set  $W = L_{1,k} - L_{2,k} + v$ . On a neighborhood where the logarithms are consistent, exactly the algebra of [Theorem 10.2](#) gives

$$\tau = 1 - e^{-v} + \sigma v, \quad \sigma = 1/L_{1,k}, \quad \tau = L_{2,k}/L_{1,k}.$$

The Lagrange-inversion calculation and coefficient identity are algebraic and remain valid over  $\mathbb{C}$ . This proves [\(132\)](#).  $\square$

This proof also shows that, for sufficiently large  $|k|$ , there is a solution near  $L_{1,k} - L_{2,k}$ : the implicit function theorem solves for  $v$  when  $\sigma$  and  $\tau$  are small. Hence a nonzero complex argument has infinitely many Lambert values.

### 18.3 Derivative and conjugation

Every regular complex branch satisfies

$$W'_k(z) = \frac{W_k(z)}{z[1 + W_k(z)]}.$$

For a positive real  $x$ , complex conjugation preserves  $we^w = x$ . Consistency with the asymptotic indexing gives

$$W_{-k}(x) = \overline{W_k(x)}.$$

Thus nonreal values occur in conjugate pairs.

*Literature note 18.2.* Under the standard DLMF convention,  $W_0$  is analytic on  $\mathbb{C} \setminus (-\infty, -1/e]$ ; the other branches are organized with cuts associated with the negative real axis and the logarithmic point at zero. Exact boundary labels require care because approaching a cut from above or below can change the branch index. The foundational paper of Corless et al. and DLMF Section 4.13 give the complete sheet structure and the standard conventions.

## 19 What the supplied archive contributes

The archive is not treated as a collection of interchangeable authorities. The foundational and reference sources establish notation and standard formulas; later papers contribute targeted approximations, bounds, algorithms, and applications. The following map records how the material was used.

| Source in the archive                                | Main contribution to this article                                                                                                                                                |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Corless, Gonnet, Hare, Jeffrey, and Knuth (1996) [5] | Modern branch notation; Maclaurin, branch-point, and all-branch asymptotic expansions; symbolic integration and numerical evaluation. This is the foundational modern reference. |
| NIST DLMF, Section 4.13 [7]                          | Canonical definitions, branch domains, coefficient conventions, asymptotic formula, and the principal-branch integral representations proved in <a href="#">Section 13</a> .     |
| Barry et al. (2000) [2]                              | Piecewise analytic approximations for both real branches, with special care at endpoints and with formulas designed as iteration starts.                                         |
| Banwell (2000) [1]                                   | Use of $W_0$ to obtain explicit current–voltage formulas in diode and bipolar-transistor circuits; see <a href="#">Theorem 16.2</a> .                                            |
| Goličnik (2012) [8]                                  | Closed forms in biochemical kinetics, including integrated Michaelis–Menten equations; see <a href="#">Theorem 16.1</a> .                                                        |
| Chatzigeorgiou (2013) [3]                            | Elementary bounds for $W_{-1}(-e^{-1-u})$ . A self-contained proof and a midpoint error estimate appear in <a href="#">Theorems 11.5</a> and <a href="#">11.6</a> .              |
| Corcino, Corcino, and Mezó (2019) [4]                | Regular, associated, and Euler continued fractions. The elementary Euler fraction and its complete partial-sum proof appear in <a href="#">Theorem 12.3</a> .                    |
| Iacono and Boyd (2017) [10]                          | Global analytic approximations and iterative refinements for the principal real branch, as well as approximation-based inequalities.                                             |
| Lóczy (2022) [13]                                    | Guaranteed and arbitrary-precision evaluation on the full real domains, especially logarithmic Newton recursions and explicit error control.                                     |
| Howard (2025) and the accompanying manuscript [9]    | Schroeder-type and transformed series, including expansions in $\log x$ , branch-point series, and comparisons of approximation regimes; see <a href="#">Section 17</a> .        |
| Lehtonen (2016) [12]                                 | A survey of ecological and evolutionary models whose implicit equations are naturally inverted by Lambert $W$ .                                                                  |
| Visser (2018) [14]                                   | Use of $W_{-1}$ in explicit approximations and bounds for the $n$ -th prime; see <a href="#">Equation (129)</a> .                                                                |
| ProductLog notebook and expository PDF/TXT files     | Computer-algebra notation, numerical cross-checks, and a broad inventory of identities. These were used as navigational aids rather than as primary proof sources.               |

Published maximum-error claims for fitted approximations depend on their stated domains and numerical searches. Rather than reproduce every fitted constant, this article concentrates on formulas whose correctness and error behavior can be proved directly: Taylor and Puiseux remainders, asymptotic orders, two-sided bounds, residual certificates, and monotone Newton iteration. The supplied Python program independently checks representative numerical behavior at high precision.

## 20 Summary and a branch-safe workflow

The Lambert function is best understood not as one mysterious formula but as a family of inverse maps. The real-variable theory can be organized around five decisions.

- Step 1. Count the real solutions before choosing a symbol.** For  $-1/e < x < 0$ , there are two values. The value in  $(-1, 0)$  is  $W_0(x)$ ; the value in  $(-\infty, -1)$  is  $W_{-1}(x)$ .
- Step 2. Use the coordinate adapted to the endpoint.** Near  $x = 0$ , use the Maclaurin series for  $W_0$ . Near the common branch point, use the signed square-root coordinate  $p = \pm\sqrt{2(1+ex)}$ . At the unbounded ends, use logarithmic coordinates.
- Step 3. Keep branch qualifications in inverse identities.** The equation  $W(we^w) = w$  is true only on the inverse interval belonging to the selected branch. This is exactly analogous to the restricted ranges used for inverse trigonometric functions.
- Step 4. For numerical work, combine an approximation with a correction and a certificate.** The branch-aware starts of [Theorem 14.4](#), logarithmic Newton iteration, and the residual bounds of [Theorem 14.1](#) form a complete real-variable procedure.
- Step 5. Expect most algebraic identities to be branch-independent, but most inequalities to be branch-dependent.** The derivative, antiderivative, contour, and asymptotic calculations begin from  $we^w = x$  and therefore apply on every regular branch. Monotonicity, curvature, signs, and bounds depend on the range of the selected inverse.

The two real branches are more symmetric than is sometimes suggested. They share the same branch point, signed Puiseux series, logarithmic asymptotic coefficient polynomials, inverse-function calculus, and monotone logarithmic Newton map. Their essential differences are global:  $W_0$  is increasing and concave on its real domain, whereas  $W_{-1}$  is decreasing, changes curvature once, and has a logarithmic singularity at zero. The exact pairing formulas of [Section 4](#) make this balance particularly clear.



## A A complete proof of Lagrange–Buermann inversion

This appendix proves the coefficient theorem used in the Maclaurin and asymptotic sections. The proof is algebraic and therefore works for formal power series; no question of numerical convergence is involved.

### A.1 Formal residues

For a Laurent series  $A(t) = \sum_{j \in \mathbb{Z}} a_j t^j$ , define

$$\text{Res}_{t=0} A(t) dt = a_{-1}.$$

Two elementary rules will be used.

**Lemma A.1** (Derivative and substitution rules). *Let  $A(t)$  be a formal Laurent series.*

- (i)  $\text{Res} A'(t) dt = 0$ .
- (ii) *If  $y = y(x) = c_1 x + c_2 x^2 + \cdots$  with  $c_1 \neq 0$ , then*

$$\text{Res}_{x=0} A(y(x)) y'(x) dx = \text{Res}_{y=0} A(y) dy. \quad (133)$$

*Proof.* For (i), the derivative of  $a_j t^j$  is  $j a_j t^{j-1}$ . A term in  $t^{-1}$  could arise only from  $j = 0$ , but its coefficient is then zero.

For (ii), linearity reduces the proof to  $A(y) = y^m$ . If  $m \neq -1$ , then

$$y(x)^m y'(x) = \frac{1}{m+1} \frac{d}{dx} y(x)^{m+1},$$

whose residue is zero by part (i); the residue of  $y^m dy$  is also zero. If  $m = -1$ , write  $y(x) = c_1 x [1 + xB(x)]$ . Then

$$\frac{y'(x)}{y(x)} = \frac{1}{x} + \frac{d}{dx} \log[1 + xB(x)],$$

so its residue is 1, exactly the residue of  $dy/y$ . This proves the substitution rule term by term.  $\square$

### A.2 Proof of the inversion formula

Suppose

$$y = x\phi(y), \quad \phi(0) \neq 0. \quad (134)$$

There is a unique formal solution  $y(x) = c_1 x + c_2 x^2 + \cdots$ : after substituting the two series, the coefficient of  $x^n$  determines  $c_n$  from the preceding coefficients, beginning with  $c_1 = \phi(0)$ .

Let  $H$  be a formal power series and  $n \geq 1$ . Coefficient extraction is a residue:

$$[x^n] H(y(x)) = \text{Res}_{x=0} \frac{H(y(x))}{x^{n+1}} dx. \quad (135)$$

From (134),

$$x = \frac{y}{\phi(y)}, \quad dx = \frac{\phi(y) - y\phi'(y)}{\phi(y)^2} dy.$$

The substitution rule of [Theorem A.1](#) therefore changes (135) into

$$[x^n] H(y(x)) = \text{Res}_{y=0} \left[ \frac{H(y)\phi(y)^n}{y^{n+1}} - \frac{H(y)\phi(y)^{n-1}\phi'(y)}{y^n} \right] dy. \quad (136)$$

Now differentiate

$$\frac{H(y)\phi(y)^n}{y^n}.$$

Its residue is zero, so

$$0 = \text{Res}_{y=0} \left[ \frac{H'(y)\phi(y)^n}{y^n} + n \frac{H(y)\phi(y)^{n-1}\phi'(y)}{y^n} - n \frac{H(y)\phi(y)^n}{y^{n+1}} \right] dy.$$

Rearranging and comparing with (136) gives

$$[x^n]H(y(x)) = \frac{1}{n} \text{Res}_{y=0} \frac{H'(y)\phi(y)^n}{y^n} dy = \frac{1}{n} [y^{n-1}] H'(y)\phi(y)^n.$$

This is (38). Taking  $H(y) = y^m$  gives

$$[x^n]y(x)^m = \frac{m}{n} [y^{n-m}] \phi(y)^n,$$

which is (39). Thus every form of Lagrange inversion used in the article has now been proved.

## B Coefficient tables and exact recurrences

The formulas in the main text are designed to generate coefficients, not merely to list them. The following tables are convenient for checking hand calculations and computer implementations.

### B.1 Maclaurin coefficients

Write

$$W_0(x) = \sum_{n \geq 1} c_n x^n, \quad c_n = \frac{(-n)^{n-1}}{n!}.$$

The first ten coefficients are

| $n$ | $c_n$    | $n$ | $c_n$         |
|-----|----------|-----|---------------|
| 1   | 1        | 6   | $-54/5$       |
| 2   | $-1$     | 7   | $16807/720$   |
| 3   | $3/2$    | 8   | $-16384/315$  |
| 4   | $-8/3$   | 9   | $531441/4480$ |
| 5   | $125/24$ | 10  | $-156250/567$ |

The last entry is intentionally left in a moderately factored rational form; the closed formula is usually more informative than a decimal expansion.

### B.2 Signed branch-point coefficients

With  $p = \pm\sqrt{2(1+ex)}$ , write

$$W = -1 + \sum_{n \geq 1} a_n p^n.$$

The first nine coefficients, generated by (51)–(52), are

| $n$ | $a_n$       | $n$ | $a_n$                   |
|-----|-------------|-----|-------------------------|
| 1   | 1           | 6   | $-221/8505$             |
| 2   | $-1/3$      | 7   | $680863/43545600$       |
| 3   | $11/72$     | 8   | $-1963/204120$          |
| 4   | $-43/540$   | 9   | $226287557/37623398400$ |
| 5   | $769/17280$ |     |                         |

The same table serves both branches; only the sign of  $p$  changes.

### B.3 Derivative polynomials

The recurrence in [Theorem 5.4](#) begins

$$\begin{aligned}
P_1(w) &= 1, \\
P_2(w) &= -(w + 2), \\
P_3(w) &= 2w^2 + 8w + 9, \\
P_4(w) &= -(6w^3 + 36w^2 + 79w + 64), \\
P_5(w) &= 24w^4 + 192w^3 + 622w^2 + 974w + 625.
\end{aligned}$$

Consequently

$$W_k^{(n)}(x) = \frac{e^{-nW_k(x)} P_n(W_k(x))}{[1 + W_k(x)]^{2n-1}}.$$

This form avoids differentiating an increasingly complicated quotient from scratch.

## C Numerical experiments and reproducibility

The archive accompanying this article contains `numerical_experiments.py`. It uses `mpmath` at 80 decimal digits to obtain reference values and then implements, independently, the approximations proved in the article:

- the signed branch-point series for both real branches;
- the logarithmic asymptotic expansion for  $W_0$  and  $W_{-1}$ ;
- the three Padé approximants of [Section 12](#);
- the two-sided  $W_{-1}$  bounds;
- the branch-safe logarithmic Newton iteration.

From the directory containing the source, run

```
python3 numerical_experiments.py --output-dir figures
```

The required packages are `mpmath`, `numpy`, and `matplotlib`; the script does not use network access. Every plotting function writes a vector PDF, so the figures remain sharp when enlarged. The reference values and defining-equation residuals are also written to `numerical_values.txt`.

Listing 1: High-precision reference values generated by the supplied script.

```
Reference values generated with mpmath at 80 decimal digits
x branch W_k(x) |W exp(W)-x|
```

```
-----
-1/e 0 -1.0 0.0
-0.35 0 -0.71663881645607385059 5.2711e-82
```

```

-0.35 -1 -1.3497172521922488334 5.2711e-82
-0.10 0 -0.11183255915896296483 0.0
-0.10 -1 -3.5771520639572972184 1.3178e-82
-0.001 -1 -9.1180064704027401213 1.2354e-83
0 0 0.0 0.0
0.1 0 0.0912765271608622643 1.3178e-82
1 0 0.567143290409783873 1.0542e-81
e 0 1.0 0.0
10 0 1.7455280027406993831 3.3735e-80
10^6 0 11.383358086140052622 1.1054e-74

```

The reference implementation is not used as a hidden proof. Its purpose is to detect transcription mistakes, illustrate the proved convergence orders, and make every graph reproducible. Analytic statements in the main text are established independently of the plots.

## D Problems with complete solutions

These exercises review the main ideas without requiring complex analysis.

### Problems

- P1.** Prove that the equation  $we^w = -1/10$  has exactly two real solutions, and locate each one in an interval of length 1.
- P2.** Let  $a \geq -1$ . Prove  $W_0(ae^a) = a$ . Let  $a \leq -1$ . Prove  $W_{-1}(ae^a) = a$ . Explain why the branch restrictions cannot be removed.
- P3.** Solve  $x + 2 \log x = 5$  for  $x > 0$ .
- P4.** Evaluate  $\int_{-1/e}^0 W_{-1}(x) dx$ .
- P5.** Show directly from the defining equation that
- $$\frac{d}{dx} e^{W_0(x)} = \frac{1}{1 + W_0(x)} \quad (x > -1/e).$$
- P6.** For  $-1/e < x < 0$ , prove  $W_0(x) + W_{-1}(x) < -2$  without differentiating either branch.
- P7.** Determine the number of positive real solutions of  $x^x = A$  for  $A > 0$ , and express them with the correct Lambert branches.
- P8.** Derive the first three nonconstant terms in the expansion of  $W_0(x)$  at  $x = 0$  by substituting  $W_0(x) = a_1x + a_2x^2 + a_3x^3 + \mathcal{O}(x^4)$  into the defining equation.
- P9.** Let  $x > e$ ,  $L = \log x$ . Prove that  $L - \log L < W_0(x) < L$ .
- P10.** Suppose an approximation  $w$  to a selected real Lambert value satisfies  $w \neq 0, -1$ . Derive one logarithmic Newton step without using  $e^w$ .

### Solutions

- S1.** Since  $-1/e < -1/10 < 0$ , [Theorem 2.1](#) gives exactly two solutions. On  $(-1, 0)$ ,  $(-1)e^{-1} < -1/10 < 0$ , so the principal solution lies in  $(-1, 0)$ . For the lower branch,

$$-4e^{-4} \approx -0.0733 > -0.1, \quad -3e^{-3} \approx -0.1494 < -0.1.$$

Because  $we^w$  is decreasing on  $(-\infty, -1)$ , the second solution lies in  $(-4, -3)$ .

**S2.** The map  $w \mapsto we^w$  is one-to-one from  $[-1, \infty)$  onto  $[-1/e, \infty)$ , so its inverse there sends  $ae^a$  back to  $a$ . This is  $W_0$ . The same argument on  $(-\infty, -1]$ , where the map is one-to-one onto  $[-1/e, 0)$ , gives the  $W_{-1}$  identity. Without the range restriction, an input in  $(-1/e, 0)$  has two preimages, so a single inverse formula cannot select both.

**S3.** Divide by 2:  $x/2 + \log x = 5/2$ . Exponentiation gives  $xe^{x/2} = e^{5/2}$ . Hence

$$\frac{x}{2}e^{x/2} = \frac{1}{2}e^{5/2}, \quad x = 2W_0\left(\frac{1}{2}e^{5/2}\right).$$

The argument is positive, so only  $W_0$  is real.

**S4.** By (29), the answer is  $-3/e$ . Directly, put  $w = W_{-1}(x)$ . Then  $w$  runs from  $-1$  to  $-\infty$ , and

$$\int we^w(1+w)dw = \left[e^w(w^2 - w + 1)\right]_{-1}^{-\infty} = -\frac{3}{e}.$$

**S5.** Differentiate and use (20):

$$\frac{d}{dx}e^{W_0(x)} = e^{W_0(x)}W_0'(x) = e^{W_0(x)}\frac{e^{-W_0(x)}}{1+W_0(x)}.$$

The factors cancel.

**S6.** Let  $\Delta = W_0(x) - W_{-1}(x) > 0$ . By (13), the sum is  $-\Delta \coth(\Delta/2)$ . Put  $y = \Delta/2$ . The function  $y \cosh y - \sinh y$  has value 0 at 0 and positive derivative  $y \sinh y$ , so  $y \coth y > 1$ . The sum is therefore less than  $-2$ .

**S7.** Put  $u = \log x$ . Then  $x^x = A$  becomes  $ue^u = \log A$ , so

$$x = \exp[W_k(\log A)].$$

If  $A > 1$ , then  $\log A > 0$  and there is one solution, from  $W_0$ . If  $A = 1$ , the positive solution is  $x = 1$ . If  $e^{-1/e} < A < 1$ , then  $-1/e < \log A < 0$  and there are two solutions, from  $W_0$  and  $W_{-1}$ . If  $A = e^{-1/e}$ , the two coalesce at  $x = e^{-1}$ . If  $0 < A < e^{-1/e}$ , there is no positive real solution.

**S8.** Expand  $e^W = 1 + W + W^2/2 + \mathcal{O}(x^3)$ . Then

$$We^W = W + W^2 + \frac{1}{2}W^3 + \mathcal{O}(x^4).$$

Substitution gives

$$a_1x + (a_2 + a_1^2)x^2 + (a_3 + 2a_1a_2 + a_1^3/2)x^3 + \mathcal{O}(x^4) = x.$$

Thus  $a_1 = 1$ ,  $a_2 = -1$ , and  $a_3 = 3/2$ .

**S9.** Since  $Le^L = Lx > x$  and  $we^w$  is increasing for  $w > 0$ , one has  $W_0(x) < L$ . Put  $a = L - \log L$ . Then

$$ae^a = x \left(1 - \frac{\log L}{L}\right) < x,$$

so the same monotonicity gives  $a < W_0(x)$ .

**S10.** Apply Newton's method to  $G(w) = w + \log|w| - \log|x|$ . Since  $G'(w) = 1 + 1/w$ ,

$$\begin{aligned} w^+ &= w - \frac{w + \log|w| - \log|x|}{1 + 1/w} \\ &= \frac{w}{1 + w} \left(1 + \log\left|\frac{x}{w}\right|\right). \end{aligned}$$

This is (107).

## E Compact formula sheet

| Topic                   | Formula and branch condition                                                                                               |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Definition              | $W_k(x)e^{W_k(x)} = x.$                                                                                                    |
| Real domains            | $W_0 : [-1/e, \infty) \rightarrow [-1, \infty)$ , increasing; $W_{-1} : [-1/e, 0) \rightarrow (-\infty, -1]$ , decreasing. |
| Derivative              | $W'_k(x) = W_k(x)/\{x[1 + W_k(x)]\}$ , with the limiting value $W'_0(0) = 1.$                                              |
| Antiderivative          | $\int W_k(x) dx = e^{W_k(x)}[W_k(x)^2 - W_k(x) + 1] + C.$                                                                  |
| Maclaurin               | $W_0(x) = \sum_{n \geq 1} (-n)^{n-1} x^n / n!,  x  < 1/e.$                                                                 |
| Branch point            | $W_{0,-1}(x) = -1 + p - p^2/3 + 11p^3/72 - \dots, p = \pm \sqrt{2(1 + ex)}.$                                               |
| Large $W_0$             | $W_0(x) = L_1 - L_2 + L_2/L_1 + \dots, L_1 = \log x, L_2 = \log L_1.$                                                      |
| Small-negative $W_{-1}$ | Same coefficient formula with $L_1 = \log(-x) < 0, L_2 = \log[-L_1].$                                                      |
| Exact branch pair       | If $\Delta = W_0 - W_{-1}$ , then $W_0 = -\Delta/(e^\Delta - 1)$ and $W_{-1} = -\Delta/(1 - e^{-\Delta}).$                 |
| Logarithmic Newton      | $w^+ = \frac{w}{1 + w} (1 + \log  x/w ).$                                                                                  |
| Residual certificate    | $ w - W_k(x)  \leq  we^w - x  / \min_I  e^t(1 + t) $ , provided the interval $I$ between $w$ and the root avoids $-1.$     |

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